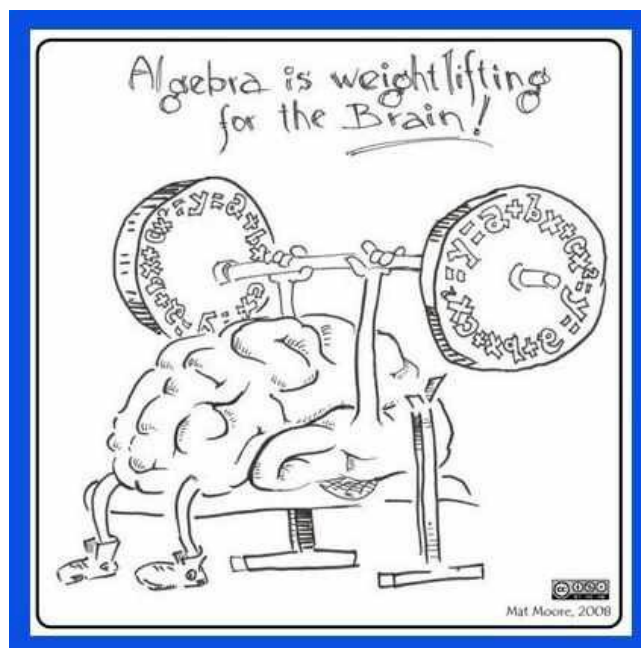


Welcome to Algebra

3AL



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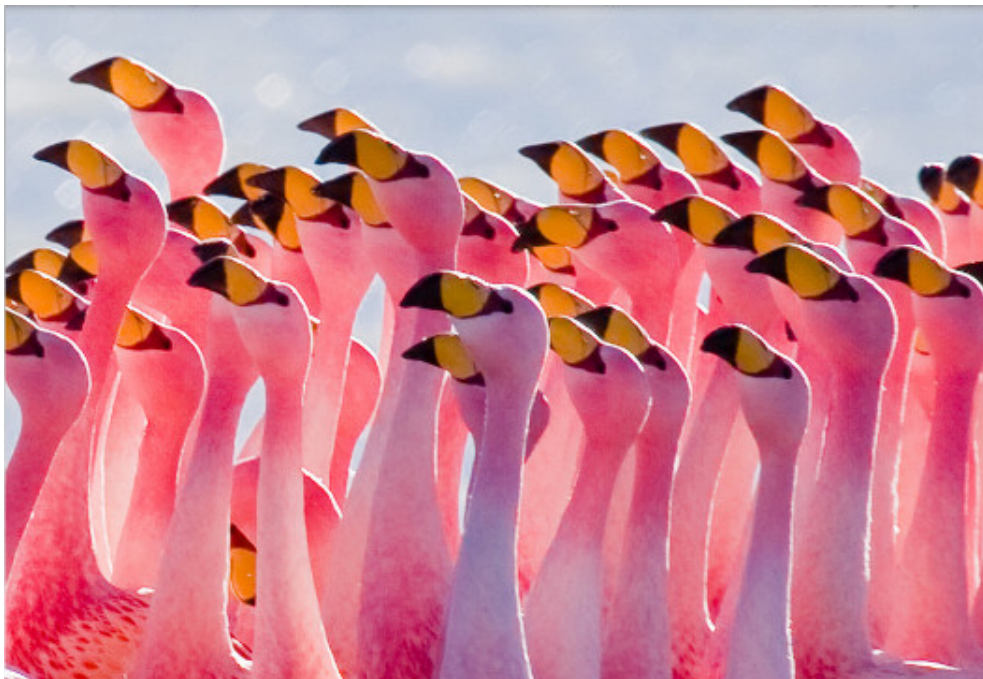
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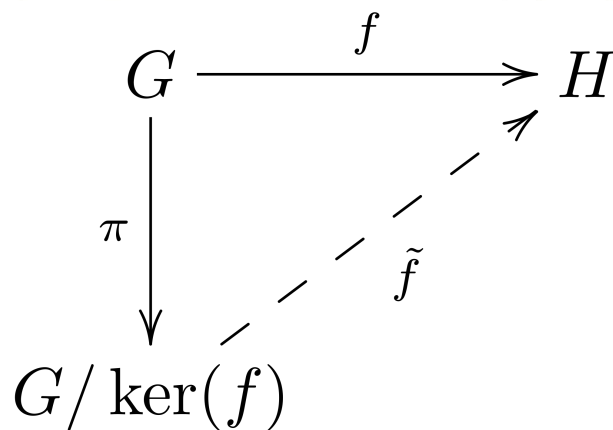
Part I

Groups



Chapter I

Lagrange's Theorem and Isomorphism Theorems



Chapter 1

Lagrange's Theorem and Isomorphism Theorems

Let us start the year recalling some results concerning to the most important algebraic structure that you learnt last year, namely, groups. Normal subgroups, cosets, factor groups and group homomorphisms will play an important role this year.

1.1 Groups

Definitions 1.1. A **group** $(G, \cdot)$ is a set G equipped with a binary operation $\cdot : G \times G \rightarrow G$ satisfying the following axioms:

- (G1) The **associative law** holds: $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ for every $a, b, c \in G$.
- (G2) There is a unity 1 for G . Then $1 \cdot a = a \cdot 1 = a$ for all $a \in G$.
- (G3) Every element of G has an inverse in G . In other words, for every $a \in G$ there is an element $b \in G$ such that $a \cdot b = b \cdot a = 1$

The group G is called **abelian** if, in addition, it satisfies

- (G4) The operation $\cdot$ is commutative: $a \cdot b = b \cdot a$ holds for every $a, b \in G$.

If G is finite, the number $|G|$ is called the **order** of G .

Although we write most groups multiplicatively, many important groups are written additively. Then the unity element is denoted 0 and is called **zero**, and the inverse of x is denoted $-x$ and is called the **negative** of x .

A subset H of a group G is called a **subgroup of G** if H is itself a group using the operation of G .

Examples 1.2.

1. The sets $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ are all abelian groups under addition. In each case the identity element is 0, and the inverse of a is $-a$. Each of them is a subgroup of the larger ones.

2. Let R denote $\mathbb{Z}_m$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$. Then the set $U(M_n(R), \cdot)$ of invertible $n \times n$ matrices over R , that is, the matrices $A \in M_n(R)$ such that $\det(A)$ has a multiplicative inverse in R , is a group called the **general linear group of degree n over R** , denoted $GL(n, R)$. More precisely:

If $R = \mathbb{Q}, \mathbb{R}$, or $\mathbb{C}$ then $GL(n, R) = \{A \in M_n(R) \mid \det(A) \neq 0\}$

$$GL(n, \mathbb{Z}) = \{A \in M_n(\mathbb{Z}) \mid \det(A) = \pm 1\}$$

$$GL(n, \mathbb{Z}_m) = \{A \in M_n(\mathbb{Z}_m) \mid \det(A) = \bar{a}, \text{ where } \gcd(a, m) = 1\}$$

The **special linear group** is the following subgroup of $GL(n, R)$:

$$SL(n, R) = \{A \in M_n(R) \mid \det(A) = 1\}$$

3. Let $n \geq 2$ be an integer. Then $\mathbb{Z}_n$ is an additive abelian group with zero $\bar{0}$ and the negative of $\bar{a}$ being $-\bar{a} = \overline{-a}$.

4. The set S_n of all permutations of $\{1, 2, \dots, n\}$ is a group under composition, called the **symmetric group of degree n** . Moreover, the **alternating group of degree n** , denoted A_n , of all **even permutations** is a subgroup of S_n .

THEOREM 1.3. Exponent Laws. *Let G be a group and $f, g, h \in G$.*

(1) $g^n g^m = g^{n+m}$ for all $n, m \in \mathbb{Z}$.

(2) $(g^n)^m = g^{nm}$ for all $n, m \in \mathbb{Z}$.

(3) If $gh = hg$, then $(gh)^n = g^n h^n$ for all $n \in \mathbb{Z}$.

(4) If $gh = gf$, then $h = f$. **Left cancellation Law**

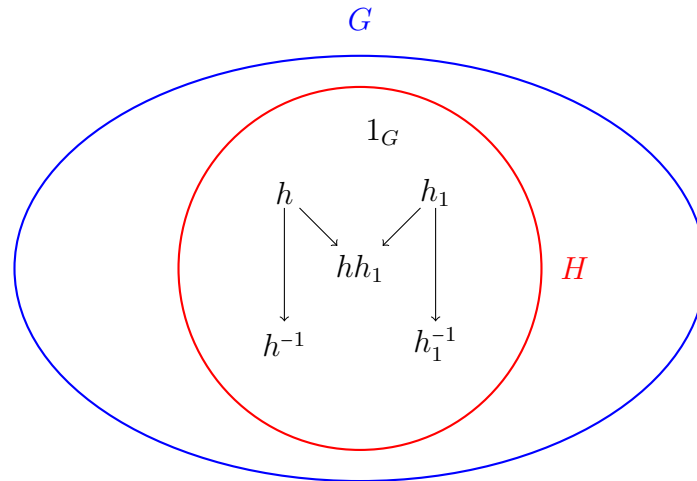
(5) If $hg = fg$, then $h = f$. **Right cancellation Law**

To find out whether a given set of a group was a subset we used the Subgroup Test:

SUBGROUP TEST. A subset H of a group G is a subgroup of G if and only if the following three conditions are satisfied.

- (1) $1_G \in H$, where 1_G is the identity element of G .
- (2) If $h \in H$ and $h_1 \in H$, then $hh_1 \in H$.
- (3) If $h \in H$, then $h^{-1} \in H$. Here h^{-1} denotes the inverse of h in G .

In this case, H has the same unity as G , and if $h \in H$, its inverse in H is the same as its inverse in G .



Definitions 1.4. Let G_1 and G_2 be two groups. A mapping $\varphi : G_1 \rightarrow G_2$ is called a **group homomorphism** if $\varphi(a \cdot_1 b) = \varphi(a) \cdot_2 \varphi(b)$, for all $a, b \in G_1$. A **monomorphism** is an injective homomorphism, while a surjective homomorphism is called an **epimorphism**. Homomorphisms which are bijectives are said to be **isomorphisms**. When an isomorphism exists from G_1 to G_2 we say that G_1 is **isomorphic to** G_2 , and we write $G_1 \cong G_2$. Isomorphisms from a group to itself are called **automorphisms**.

The set consisting of all automorphisms of G , denoted $\text{Aut}(G)$, is a group under composition called the **group of automorphisms of** G .

An example of a group homomorphism is the **determinant map**

$$\det : GL_n(\mathbb{R}) \rightarrow \mathbb{R}^* = \mathbb{R} - \{0\}, \quad A \mapsto \det(A)$$

It is a homomorphism because $\det(AB) = \det(A)\det(B)$, for all matrices A and B , and $\det(AB) \neq 0$. Moreover, the determinant map is an epimorphism, since for any $r \in \mathbb{R}^*$ we have that $r = \det(rE_{11} + E_{22} + \dots + E_{nn})$ where E_{ii} is the matrix having 1 in position (i, i) and 0 elsewhere. The determinant map is not a monomorphism since, for example, $\det(2E_{11} + E_{22} + \dots + E_{nn}) = \det(E_{11} + 2E_{22} + \dots + E_{nn}) = 2$ and the matrices $2E_{11} + E_{22} + \dots + E_{nn}$ and $E_{11} + 2E_{22} + \dots + E_{nn}$ are different.

1.2 Dihedral groups

Let $n \geq 2$ be an integer, the **dihedral group** D_n is the group of order $2n$ presented as $D_n = \{1, a, a^2, \dots, a^{n-1}, b, ba, ba^2, \dots, ba^{n-1}\}$, where $\text{o}(a) = n$, $\text{o}(b) = 2$, and $aba = b$.

Notice that a and b are the generators of D_n . We can compute any product in D_n using that $\text{o}(a) = n$, $\text{o}(b) = 2$ and $aba = b$. Moreover, from $aba = b$ and using induction we can obtain that $a^k ba^k = b$ for all $k \in \mathbb{Z}$. Using that $a^n = 1$ we get that $a^k b = ba^{-k} = ba^{n-k}$ for all $k \in \mathbb{Z}$. It implies $\text{o}(ba^k)^2 = 1$ for all $k \in \mathbb{Z}$ since

$$(ba^k)^2 = (ba^k)(ba^k) = b(a^k b)a^k = b(ba^{n-k})a^k = b^2 a^n = 1$$

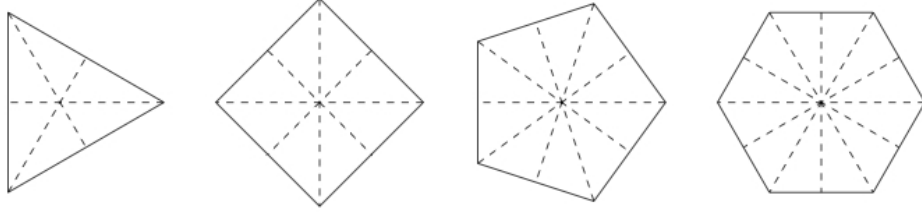
Using these new formulas we can fill in the Cayley table for D_n . For example, the Cayley table for D_4 , which is called the **octic group**, is as follows:

D_4	1	a	a^2	a^3	b	ba	ba^2	ba^3
1	1	a	a^2	a^3	b	ba	ba^2	ba^3
a	a	a^2	a^3	1	ba^3	b	ba	ba^2
a^2	a^2	a^3	1	a	ba^2	ba^3	b	ba
a^3	a^3	1	a	a^2	ba	ba^2	ba^3	b
b	b	ba	ba^2	ba^3	1	a	a^2	a^3
ba	ba	ba^2	ba^3	b	a^3	1	a	a^2
ba^2	ba^2	ba^3	b	ba	a^2	a^3	1	a
ba^3	ba^3	b	ba	ba^2	a	a^2	a^3	1

Notice that D_3 is isomorphic to S_3 because $D_3 = \{1, a, a^2, b, ba, ba^2\}$, $\text{o}(a) = 3$, $\text{o}(b) = 2$, and $aba = b$. In S_3 make $a = (1\ 2\ 3)$ and $b = (1\ 2)$.

If $n = 2$, then D_2 is the Klein group because $D_2 = \{1, a, b, ab\}$, $\text{o}(a) = 2$, $\text{o}(b) = 2$, and $aba = b$. From $\text{o}(a) = 2$ we obtain that $a^{-1} = a$, which implies that D_2 is abelian ($ba = a^{-1}b = ab$).

If $n \geq 3$, as you saw in 2IA, the dihedral group D_n can be seen as the group of motions of a regular n -gon, which is a plane figure with n vertices evenly placed on a circle.



By a motion of it we mean a permutation of its vertices that can be realized by either a rotation or a symmetry. More precisely, we can reflect the polygon across each of the dotted lines, or rotated by any multiple of $2\pi/n$ radians around the center. Such these motions bring the polygon back to itself. In terms of permutations $a = (1\ 2\ 3\ \dots\ n)$ (the clockwise rotation of $2\pi/n$ radians) and $b = (1\ n-1)(2\ n-2)(3\ n-3)\dots$ (the rotation of π radians about a line through the vertex n and the center of the n -gon).

Let us now focus our attention on the octic group

$$D_4 = \{1, a, a^2, a^3, b, ba, ba^2, ba^3\}, \quad o(a) = 4, o(b) = 2, aba = b$$

To get more familiar with this group, let us prove that its centre is $Z(D_4) = \{1, a^2\}$. Notice that we need to show that the only elements of D_4 which commute with any other element in D_4 are 1 and a^2 . It is clear that 1 commutes with all the elements in D_4 .

We will use that $a^k b = ba^{-k} = ba^{4-k}$ for all $k \in \mathbb{Z}$. Let $n \in \{0, 1, 2, 3\}$ and let us show that $ba^n \notin Z(D_4)$. From

$$a(ba^n) = (ab)a^n = (ba^3)a^n = ba^{n+3} \neq ba^{n+1} = (ba^n)a,$$

we can see that ba^n does not commute with a , which implies that $ba^n \notin Z(D_4)$ for $n \in \{0, 1, 2, 3\}$. On the other hand, from

$$ab = ba^3 \neq ba, \quad a^3b = ba \neq ba^3$$

we conclude that $a, a^3 \notin Z(D_4)$. We have now proved that $Z(D_4) \subseteq \{1, a^2\}$. But notice that a^2 commutes with both generators a and b ($a^2a = a^3 = aa^2$, $a^2b = ba^2$), which says that $a^2 \in Z(D_4)$ and so $Z(D_4) = \{1, a^2\}$.

1.3 Cosets

In this chapter we will prove Lagrange's Theorem, one of the most important results in (finite) group theory. To do so, we first need to introduce the required machinery: cosets. The notion of a coset was introduced by Galois in 1930 but was G. A. Miller the one who name them cosets.

Definition 1.5. Let H be a subgroup of a group G and $a \in G$. The set $Ha = \{ha \mid h \in H\}$ is called the **right coset** of H in G **generated** by a . Similarly, $aH = \{ah \mid h \in H\}$ is said to be the **left coset** of H in G **generated** by a .

Observe that in **additive notation the left and right cosets** are:

$$H + a = \{h + a \mid h \in H\}, \quad a + H = \{a + h \mid h \in H\}$$

It is important to notice that **every coset in a group G is a subset of G . Cosets are not elements in G !**

Example 1.6. Let $G = S_3$ and $H = \{\varepsilon, (1\ 3)\}$. Find all the left cosets of H in G .

Solution: Recall that $S_3 = \{\varepsilon, (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$. You can easily check that

$$\begin{aligned} \varepsilon H &= H = (1\ 3)H, \\ (1\ 2)H &= \{(1\ 2), (1\ 3\ 2)\} = (1\ 3\ 2)H, \\ (2\ 3)H &= \{(2\ 3), (1\ 2\ 3)\} = (1\ 2\ 3)H. \end{aligned}$$

Remark 1.7. Let $\sigma = (1\ 2\ 3)$ and $\tau = (1\ 2)$, then you can easily check that σ and τ generates S_3 . Moreover, S_3 can be presented as follows:

$$S_3 = \{\varepsilon, \sigma, \sigma^2, \tau, \tau\sigma, \tau\sigma^2\},$$

where $\sigma^3 = \tau^2 = \varepsilon$ and $\sigma\tau\sigma = \tau$. From now on, we will write S_3 as indicated above.

Example 1.8. Find all the right cosets of the subgroup $4\mathbb{Z}$ in the additive group $\mathbb{Z}$.

Solution: Given $a \in \mathbb{Z}$ the right coset generated by a is $4\mathbb{Z} + a$. But

$$4\mathbb{Z} + a = a + 4\mathbb{Z} = \bar{a} \quad \text{in } \mathbb{Z}_4.$$

We only have four right cosets, namely, the four residue classes modulo 4:

$$4\mathbb{Z} + 0 = \{\dots, -12, -8, -4, 0, 4, 8, 12, \dots\}$$

$$4\mathbb{Z} + 1 = \{\dots, -11, -7, -3, 1, 5, 9, 13, \dots\}$$

$$4\mathbb{Z} + 2 = \{\dots, -10, -6, -2, 2, 6, 10, 14, \dots\}$$

$$4\mathbb{Z} + 3 = \{\dots, -9, -5, -1, 3, 7, 11, 15, \dots\}$$

Example 1.9. Consider the octic group $D_4 = \{1, a, a^2, a^3, b, ba, ba^2, ba^3\}$, where $\text{o}(a) = 4$, $\text{o}(b) = 2$ and $aba = b$. Find the cosets of $Z(D_4) = \{1, a^2\}$.

Solution: Let us write $Z = Z(D_4)$ for short. Notice that left and right cosets coincide because Z is the centre and recall that elements in the centre commutes with every element in the group.

$$Z1 = Z$$

$$Za = \{a, a^3\}$$

$$Za^2 = \{a^2, a^4 = 1\} = Z$$

$$Za^3 = \{a^3, a^5 = a\} = Za$$

$$Zb = \{b, a^2b = ba^2\} = \{b, ba^2\}$$

$$Z(ba) = \{ba, a^2(ba)\} = \{ba, ba^3\}$$

$$Z(ba^2) = \{ba^2, a^2(ba^2)\} = \{ba^2, ba^4 = b\} = \{ba^2, b\} = Zb$$

$$Z(ba^3) = \{ba^3, a^2(ba^3)\} = \{ba^3, ba^5\} = \{ba^3, ba\} = Z(ba)$$

Thus, the different cosets are Z , Za , Zb and $Z(ba)$.

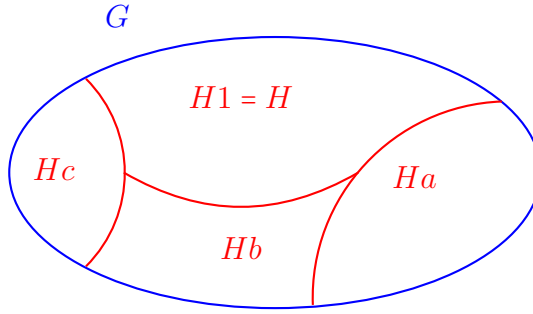
Some important properties of cosets

Let $a \in G$ and H a subgroup of G . Then:

1. $H1 = H = 1H$, which means that H is a right and left coset of itself.

2. It is obvious that $a \in Ha$ and $a \in aH$ since $a = 1 \cdot a = a \cdot 1$ for $1 \in H$.

- 3.** If G is abelian, then $Ha = aH$ for all $a \in G$ and all subgroups H of G .
- 4.** **3** might not be true if G is not abelian.
- 5.** If $a \in Hb$, then $Ha = Hb$.
- 6.** $Ha = Hb$ if and only if $ab^{-1} \in H$.
- 7.** $Ha = H$ if and only if $a \in H$.
- 8.** There is a bijection from H to Ha . In particular, if H is finite, then any coset Ha has the same number of elements as H and therefore all the cosets have the same number of elements.
- 9.** The family of all the cosets of Ha , as a ranges over G , is a partition of G . It means that $G = \cup_{a \in G} Ha$, and $Ha \neq Hb$ implies $Ha \cap Hb = \emptyset$.



Observe that **6** and **7** in additive notation become

$$H + a = H + b \Leftrightarrow a - b \in H; \quad H + a = H \Leftrightarrow a \in H$$

For instance, $6\mathbb{Z} + 7 = 6\mathbb{Z} + 1$ since $7 - 1 = 6 \in 6\mathbb{Z}$.

Proof. Properties **1**, **2** and **3** trivially hold.

- 4.** Take $G = S_3$ and $H = \{\varepsilon, (1\ 3)\}$. Then one can easily check that

$$(1\ 2)H = \{(1\ 2), (1\ 3\ 2)\} \neq \{(1\ 2), (1\ 2\ 3)\} = H(1\ 2)$$

5. To show that the cosets Ha and Hb are equal, we must show that they are equal sets. Suppose that $a \in Hb$. Then there exists $h \in H$ such that $a = hb$. If $x \in Ha$, then $x = h_1a$ for some $h_1 \in H$. From $a = hb$ we get that

$x = h_1a = h_1(hb) = (hh_1)b \in Hb$. This proves that $Ha \subseteq Hb$. The proof of the containment $Hb \subseteq Ha$ is similar (use now that $b = h^{-1}a$).

6. If $ab^{-1} \in H$, then $ab^{-1} = h$ and so $a = hb \in Hb$. An application of **5** gives that $Ha = Hb$. If $Ha = Hb$, then $a \in Hb$ so $a = hb$ for some $h \in H$. This gives that $ab^{-1} = h \in H$, as desired.

7 follows from **6** by taking $b = 1$.

8. Define $\psi : H \rightarrow Ha$ by $\psi(h) = ha$. Clearly, ψ is surjective because every element of Ha is of the form ha for some $h \in H$ and so $ha = \psi(h)$. Lastly, if $\psi(h_1) = \psi(h_2)$, then $h_1a = h_2a$ and applying the cancellation law in G we get that $h_1 = h_2$, which proves that ψ is injective.

9. We first need to show that any two cosets, say Ha and Hb , are either disjoint or equal. If they are disjoint, then we are done. Otherwise, let $x \in Ha \cap Hb$. Then $x = h_1a = h_2b$ for some $h_1, h_2 \in H$. Thus solving for a , we obtain $a = (h_1^{-1}h_2)b \in Hb$. It follows from **5** that $Ha = Hb$.

Next, we need to check that G is the union of all the cosets. In order to do so, we must show that every element $g \in G$ is in one of the cosets of H . But this clearly holds, since $g = 1 \cdot g \in Hg$.

Thus, the family of all the cosets of H is a partition of G . □

1.4 Lagrange's Theorem

Now we are in a position to prove Lagrange's Theorem.

LAGRANGE'S THEOREM.

Let H be a subgroup of a finite group G . Then the order of G is a multiple of the order of H .

PROOF. Let $Ha_1, Ha_2, \dots, Ha_k$ be the distinct cosets of H in G . Then $G = Ha_1 \cup Ha_2 \cup \dots \cup Ha_k$, which is a disjoint union by Property **9**. Next, using Property **8** we have that $|Ha_i| = |H|$ for each i , so

$$|G| = |Ha_1| + \dots + |Ha_k| = |H| + \dots + |H| = k|H|,$$

which shows that the order of G is a multiple of the order of H , as desired. $\square$

A few applications of Lagrange's Theorem follow.

COROLLARY 1.10. *The order of any element of a finite group divides the order of the group.*

PROOF. Let G be a finite group and $g \in G$. By definition the order $\text{o}(g)$ of g coincides with the order of the cyclic subgroup $\langle g \rangle$ of G generated by g . The result now follows by applying Lagrange's Theorem to $\langle g \rangle$ and G . $\square$

COROLLARY 1.11. *Every group of prime order is cyclic.*

PROOF. Let p be a prime and G a group of order p . Take $g \in G$ with $g \neq 1$ and consider the subgroup $\langle g \rangle$ of G , which is non-trivial because $g \neq 1$. By Lagrange's Theorem we have that $|\langle g \rangle| \mid p$, and so $|\langle g \rangle| = p$, that is, $G = \langle g \rangle$. $\square$

Last year, you saw that every group of order $4 = 2^2$ is either cyclic or isomorphic to the Klein group. We can now use Lagrange's Theorem to prove an analogous result for an arbitrary prime p .

COROLLARY 1.12. *If $|G| = p^2$ where p is a prime, then either G is cyclic or $g^p = 1$ for every element g in G .*

PROOF. Assume that G is not cyclic, and let $g \in G$. Write $H = \langle g \rangle$ for the cyclic subgroup of G generated by g . Then by Corollary 1.11 the order of H , namely, $o(g) = |H|$ divides the order of G , i.e., p^2 . From here we derive that $o(g) = 1, p$, or p^2 . But $o(g) \neq p^2$ because G is not cyclic, so $o(g)$ is either 1 or p . In any case, $g^p = 1$, concluding the proof. $\square$

Let G be a group and H a subgroup of G , the **index of H in G** is the number of cosets of H in G . We denote it by $|G : H|$. If G is finite, from the proof of Lagrange's Theorem, we know that the number of elements in G is equal to the number of elements in H , multiplied by the number of cosets of H in G . In other words:

$$|G : H| = \frac{\text{order of } G}{\text{order of } H} = \frac{|G|}{|H|}$$

Note that H can be of finite index in G even if both H and G are infinite. For example, $H = 4\mathbb{Z}$ has index 4 in $G = \mathbb{Z}$ and both $4\mathbb{Z}$ and $\mathbb{Z}$ are infinite.

1.5 Normal subgroups

Let H be a subgroup of a group G and $a \in G$, we have already seen that the left and right cosets aH and Ha might be different. We say that H is a **normal subgroup** of G , or that H is **normal in G** , if $gH = Hg$ for all $g \in G$. We write $H \triangleleft G$.

Examples 1.13.

1. Let G be a group. Then $\{1\} \triangleleft G$ and $G \triangleleft G$ since $g\{1\} = \{g\} = \{1\}g$, $gG = G = Gg$, for all $g \in G$. The centre $Z(G)$ of G is always a normal subgroup of G .

2. Every subgroup of an abelian group is normal.

3. Let $\varphi : G \rightarrow G'$ be a group homomorphism. Then $\ker \varphi \triangleleft G$.

- 4.** Let $G = S_3 = \{\varepsilon, \sigma, \sigma^2, \tau, \tau\sigma, \tau\sigma^2\}$, where $\sigma^3 = \tau^2 = \varepsilon$ and $\sigma\tau\sigma = \tau$. Write $H = \{\varepsilon, \tau\}$ and $K = \{\varepsilon, \sigma, \sigma^2\}$. Then one can check that $K \triangleleft S_3$ but H is not normal in S_3 . In fact:

$$\begin{aligned} H\sigma &= \{h\sigma \mid h \in H\} = \{\varepsilon\sigma, \tau\sigma\} = \{\sigma, \tau\sigma\}, \\ \sigma H &= \{\sigma h \mid h \in H\} = \{\sigma\varepsilon, \sigma\tau\} = \{\sigma, \tau\sigma^2\}, \end{aligned}$$

since $\sigma\tau = \sigma^2\tau\sigma = \sigma^3\tau\sigma^2 = \tau\sigma^2$. Thus $H\sigma \neq \sigma H$, and H is not normal in S_3 . On the other hand, $K \triangleleft S_3$ because

$$\begin{aligned} \alpha K &= K = K\alpha, \text{ for all } \alpha \in K, \\ \tau K &= \{\tau, \tau\sigma, \tau\sigma^2\} = K\tau, \\ \tau\sigma K &= \{\tau\sigma, \tau\sigma^2, \tau\} = K\tau\sigma, \\ \tau\sigma^2 K &= \{\tau\sigma^2, \tau, \tau\sigma\} = K\tau\sigma^2. \end{aligned}$$

An easy way to test whether a subgroup is normal follows.

THEOREM 1.14. NORMALITY TEST.

The following conditions are equivalent for a subgroup of a group G .

- (1) H is normal in G .
- (2) $gHg^{-1} \subseteq H$ for all $g \in G$.
- (3) $gHg^{-1} = H$ for all $g \in G$.

PROOF. (1) $\Rightarrow$ (2). Let $x \in gHg^{-1}$, say $x = ghg^{-1}$ for some $h \in H$. Then $gh \in gH = Hg$ by (1), and so $gh = h_1g$ for some $h_1 \in H$. But then $x = ghg^{-1} = h_1gg^{-1} = h_1 \in H$, which proves (2).

(2) $\Rightarrow$ (3). Take $g \in G$ then applying (2) with g^{-1} we get that $g^{-1}Hg \subseteq H$. Therefore:

$$H = g(g^{-1}Hg)g^{-1} \subseteq gHg^{-1} \stackrel{(2)}{\subseteq} H \Rightarrow H = gHg^{-1},$$

proving (3).

(3) $\Rightarrow$ (1). Given $g \in G$, we have to prove that $gH = Hg$. We will show that $gH \subseteq Hg$. The other containment can be proved in a similar way by replacing g by g^{-1} . If $x \in gH$ from $gHg^{-1} = H$ we have that $xg^{-1} \in H$, and so $x = (xg^{-1})g \in Hg$. $\square$

The most useful part of the Normality Test is (2). Let us see an example.

Example 1.15. The special linear group $SL(2, \mathbb{R})$ is a normal subgroup of the general linear group $GL(2, \mathbb{R})$.

Solution: Let $H = SL(2, \mathbb{R})$ and $G = GL(2, \mathbb{R})$. Recall that

$$GL(2, \mathbb{R}) = \{A \in M_2(\mathbb{R}) : \det(A) \neq 0\}$$

$$SL(2, \mathbb{R}) = \{A \in M_2(\mathbb{R}) : \det(A) = 1\}$$

We will prove that H satisfies (2) in the Normality Test. Given $A \in G$, we need to show that $AHA^{-1} \subseteq H$. If $B \in H$, then

$$\det(ABA^{-1}) = \det(A)\det(B)\frac{1}{\det(A)} = \det(B) = 1 \Rightarrow ABA^{-1} \in H,$$

as desired.

1.6 Factor groups

Let G be a group and H a subgroup of G . We can combine cosets, via the **coset operation** as follows: the coset of a , operated with the coset of b is defined to be the coset of ab . In symbols,

$$Ha \cdot Hb = H(ab)$$

This definition seems to be quite innocent but it is not that simple! The difficulty arises from the fact that the result of operate two cosets Ha and Hb might not be uniquely defined. Recall that Ha may be the same coset as Hc (it happens if $c \in Ha$), and similarly, Hb may be the same coset as Hd . Therefore, $Ha \cdot Hb$ should be the same as $Hc \cdot Hd$. Unfortunately, it may happen that $H(ab)$ is not the same coset as $H(cd)$. Graphically,

$$\begin{array}{ccc} Ha \cdot Hb & = & H(ab) \\ \parallel & & \parallel \\ Hc \cdot Hd & = & H(cd) \end{array}$$

For example, if $G = S_3$ and $H = \{\varepsilon, (1\ 2)\}$, then

$$H(1\ 3) = \{(1\ 3), (1\ 3\ 2)\} = H(1\ 3\ 2)$$

$$H(2\ 3) = \{(2\ 3), (1\ 2\ 3)\} = H(1\ 2\ 3)$$

and yet $H(1\ 3)(2\ 3) = H(1\ 3\ 2) \neq H = H\varepsilon = H(1\ 3\ 2)(1\ 2\ 3)$.

Therefore, coset multiplication does not work as an operation on the cosets of $H = \{\varepsilon, (1\ 2)\}$ in S_3 . Why not? What did it go wrong? Well ...

Although H is a subgroup, H is not a normal subgroup of S_3 . So then the next questions are “what if H were a normal subgroup?” “Would coset operation work?” Good news! ☺☺ Check out the following theorem!

THEOREM 1.16. *Let H be a normal subgroup of a group G . If $Ha = Hc$ and $Hb = Hd$ then $H(ab) = H(cd)$.*

PROOF. Assume that $Ha = Hc$ and $Hb = Hd$. Then $a \in Hc$ and $b \in Hd$, hence $a = h_1c$ and $b = h_2d$ for some $h_1, h_2 \in H$. Thus,

$$ab = (h_1c)(h_2d) = h_1(ch_2)d$$

But $ch_2 \in cH = Hc$ because H is normal, and so $ch_2 = h_3c$ for some $h_3 \in H$. Returning to ab we get that

$$ab = h_1(ch_2)d = h_1(h_3c)d = (h_1h_3)(cd) \in H(cd)$$

An application of Cosets Property **5** gives that $H(ab) = H(cd)$. □

Let G be a group and H a normal subgroup of G . Let G/H be the set consisting of all the cosets of H in G . Put it this way, if $Ha, Hb, Hc, \dots$ are the cosets of H in G , then

$$G/H = \{Ha, Hb, Hc, \dots\} = \{Hg \mid g \in G\}$$

We have just proved that coset operation is a valid operation on G/H . In our next result, we prove that G/H is indeed a group under coset operation.

THEOREM 1.17. *Let G be a group and H a normal subgroup of G . Then G/H is a group under coset operation: $Ha \cdot Hb = H(ab)$*

PROOF. 1. Coset operation is associative:

$$\begin{aligned} Ha \cdot (Hb \cdot Hc) &= Ha \cdot H(bc) = H(a(bc)) = H((ab)c) = H(ab) \cdot Hc = \\ &= (Ha \cdot Hb) \cdot Hc \end{aligned}$$

2. The unity for G/H is H :

$$Ha \cdot H = Ha \cdot H1 = H(a1) = H(1a) = H1 \cdot Ha = H \cdot Ha$$

3. The inverse of Ha is Ha^{-1} :

$$Ha \cdot Ha^{-1} = H(aa^{-1}) = H1 = H, \quad Ha^{-1} \cdot Ha = H(a^{-1}a) = H1 = H$$

□

The group G/H is said to be the **factor group**, or **quotient group** of G by H . Note that $|G/H| = |G : H|$. Recall that $|G : H|$ denotes the number of distinct cosets of H in G , and it is called the **index of H in G** . An important map is the **coset map** $\pi : G \rightarrow G/H$ given by $\pi(a) = Ha$. The coset map is a **group epimorphism** with kernel $\ker \pi = H$.

THEOREM 1.18. *Let H be a normal subgroup of a group G .*

- (1) *If G is abelian, then G/H is abelian.*
- (2) *If $G = \langle g \rangle$ is cyclic, then G/H is also cyclic: $G/H = \langle Hg \rangle$.*
- (3) *If $H \subseteq Z(G)$ satisfies that G/H is cyclic, then G is abelian.*

PROOF. Tutorials! □

Example 1.19. Let $\mathbb{Z}$ be the additive group of the integers, and let $\langle 6 \rangle = 6\mathbb{Z}$ be the cyclic subgroup of $\mathbb{Z}$ consisting of all the multiples of 6. Notice that $6\mathbb{Z}$ is a normal subgroup of $\mathbb{Z}$ because $\mathbb{Z}$ is abelian (so all its subgroups are normal). Therefore, we can consider the factor group $\mathbb{Z}/6\mathbb{Z}$. The elements of this group are the cosets of $6\mathbb{Z}$ in $\mathbb{Z}$, namely:

$$6\mathbb{Z} + 0 = \{\dots, -18, -12, -6, 0, 6, 12, 18, \dots\}$$

$$6\mathbb{Z} + 1 = \{\dots, -17, -11, -5, 1, 7, 13, 19, \dots\}$$

$$6\mathbb{Z} + 2 = \{\dots, -16, -10, -4, 2, 8, 14, 20, \dots\}$$

$$6\mathbb{Z} + 3 = \{\dots, -15, -9, -3, 3, 9, 15, 21, \dots\}$$

$$6\mathbb{Z} + 4 = \{\dots, -14, -8, -2, 4, 10, 16, 22, \dots\}$$

$$6\mathbb{Z} + 5 = \{\dots, -13, -7, -1, 5, 11, 17, 23, \dots\}$$

These are all the different cosets of $6\mathbb{Z}$. Observe that $6\mathbb{Z} + 6 = 6\mathbb{Z} + 0$ (since $6 \in 6\mathbb{Z} + 0$), $6\mathbb{Z} + 7 = 6\mathbb{Z} + 1$ (since $7 \in 6\mathbb{Z} + 1$), $6\mathbb{Z} + 8 = 6\mathbb{Z} + 2$, etc.

Next, the operation in $\mathbb{Z}$ is $+$, so we use $+$ for the coset operation, as well. But nothing has changed except for the name: $(6\mathbb{Z} + a) + (6\mathbb{Z} + b) = 6\mathbb{Z} + (a + b)$. To ease our notation, we write the coset in the following shorter form:

$$\bar{0} = 6\mathbb{Z} + 0 \quad \bar{1} = 6\mathbb{Z} + 1 \quad \bar{2} = 6\mathbb{Z} + 2 \quad \bar{3} = 6\mathbb{Z} + 3 \quad \bar{4} = 6\mathbb{Z} + 4 \quad \bar{5} = 6\mathbb{Z} + 5$$

Then $\mathbb{Z}/6\mathbb{Z} = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}\}$, and its Cayley table is:

$+$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$
$\bar{0}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$
$\bar{1}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$	$\bar{0}$
$\bar{2}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$	$\bar{0}$	$\bar{1}$
$\bar{3}$	$\bar{3}$	$\bar{4}$	$\bar{5}$	$\bar{0}$	$\bar{1}$	$\bar{2}$
$\bar{4}$	$\bar{4}$	$\bar{5}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$
$\bar{5}$	$\bar{5}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$

Notice that the factor group $\mathbb{Z}/6\mathbb{Z}$ is isomorphic to $\mathbb{Z}_6$. In general, $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$.

Example 1.20. Consider the octic group $D_4 = \{1, a, a^2, a^3, b, ba, ba^2, ba^3\}$, where $o(a) = 4$, $o(b) = 2$ and $aba = b$. Find the Cayley table of the factor group $D_4/Z(D_4)$ and conclude that $D_4/Z(D_4)$ is isomorphic to the Klein group K_4 .

Solution: Write $Z = Z(D_4)$ for short. From Example 1.9 we know that $D_4/Z = \{Z, Za, Zb, Z(ba)\}$. Keeping in mind how multiplication works in factor groups and the Cayley table of D_4 displayed in § 1.2 we obtain the following:

D_4/Z	Z	Za	Zb	$Z(ba)$
Z	Z	Za	Zb	$Z(ba)$
Za	Za	Z	$Z(ba)$	Zb
Zb	Zb	$Z(ba)$	Z	Za
$Z(ba)$	$Z(ba)$	Zb	Za	Z

From the table we can see that D_4/Z is not a cyclic group and has order 4, so $D_4/Z \cong K_4$. Remember that (up to isomorphisms) there are only two groups of order 4: the cyclic group C_4 and the Klein group K_4 .

1.7 The Correspondence Theorem

The goal in this section is to study the subgroups of a factor group G/K . We will show that there is a correspondence between the set of subgroups of G that contains K and the subgroups of G/K . Moreover, if we are able to determine all the subgroups in one of the sets, then we will be also able to easily compute the subgroups in the other set.

To motivate and also understand how this correspondence works, we take $K \triangleleft G$ and H a subgroup of G such that $K \subseteq H$. Notice that $K \triangleleft H$ so it makes sense to consider the factor group H/K , defined as

$$H/K = \{Kh \mid h \in H\}$$

It is obvious that H/K is a subgroup of G/K (just remember the coset operation and do not forget that H is a subgroup of G). We are going to prove that all the subgroups of G/K arises as this way.

Let us start with a very simple but useful result.

LEMMA 1.21. *Let $K \subseteq H \subseteq G$ be groups with $K \triangleleft G$. If $K \subseteq H_1 \subseteq G$, then $H \subseteq H_1$ if and only if $H/K \subseteq H_1/K$.*

PROOF. It is clear that $H \subseteq H_1$ implies $H/K \subseteq H_1/K$. Conversely, assume that $H/K \subseteq H_1/K$ and take $h \in H$. Then $Kh \in H/K \subseteq H_1/K$, so we can find $h_1 \in H_1$ such that $Kh = Kh_1$. This is equivalent to say that $hh_1^{-1} \in K \subseteq H_1$. Therefore $h = (hh_1^{-1})h_1 \in H_1$. $\square$

THEOREM 1.22. CORRESPONDENCE THEOREM. *Let $K \triangleleft G$ be groups and define a map*

$$\Theta : \{H \mid H \text{ is a subgroup of } G, K \subseteq H\} \rightarrow \{\mathcal{H} \mid \mathcal{H} \text{ is a subgroup of } G/K\}$$

by $\Theta(H) = H/K$. Then:

- (1) *If $\mathcal{H}$ is a subgroup of G/K , then $\mathcal{H} = H/K$ where $H = \{g \in G \mid Kg \in \mathcal{H}\}$.*
- (2) *The map Θ is a bijection.*

Let H , H_1 and H_2 be subgroups of G containing K . Then:

- (3) Θ preserves containment: $H \subseteq H_1$ if and only if $H/K \subseteq H_1/K$.
- (4) Θ preserves normality: $H \triangleleft H_1$ if and only if $H/K \triangleleft H_1/K$.
- (5) Θ preserves intersections: $(H_1/K) \cap (H_2/K) = (H_1 \cap H_2)/K$.
- (6) Θ preserves products: $(H_1/K)(H_2/K) = (H_1H_2)/K$ if H_1H_2 is a subgroup of G .

PROOF. (1). Let $\mathcal{H}$ be a subgroup of G/K , and define $H = \{g \in G \mid Kg \in \mathcal{H}\}$. Notice that $K \subseteq H$ since for any $k \in K$ we have that $Kk = K \in \mathcal{H}$ because $\mathcal{H}$ is a subgroup of G/K . Moreover, we claim that H is a subgroup of G . In fact, if $h, h' \in H$, then $Kh, Kh' \in \mathcal{H}$ and so

$$\begin{aligned} K(hh') &= (Kh)(Kh') \in \mathcal{H}, \\ Kh^{-1} &= (Kh)^{-1} \in \mathcal{H} \end{aligned}$$

Thus $hh' \in H$ and $h^{-1} \in H$ and the Subgroup Test allows us to conclude that H is a subgroup of G .

We can now consider $H/K \stackrel{?}{=} \mathcal{H}$.

$\mathcal{H} \subseteq H/K$ because $Kg \in \mathcal{H}$ implies (by definition of H) that $g \in H$, and so $Kg \in H/K$. Conversely, if $Kg \in H/K$, then $Kg = Kh$ for some $h \in H$. From $K \subseteq H$ we obtain that $g \in Kh \subseteq Hh \subseteq H$ and so $Kg \in \mathcal{H}$. It shows that $H/K \subseteq \mathcal{H}$ and hence $H/K = \mathcal{H}$, as required.

(2). Notice that Θ is surjective by (1). If $\theta(H) = \theta(H_1)$, that is, $H/K = H_1/K$. Then from Lemma 1.22 we can conclude that $H = H_1$. It shows that Θ is injective.

(3) follows from Lemma 1.22.

(4). Assume that $H/K \triangleleft H_1/K$ and let us show that $H \triangleleft H_1$. Given $h \in H$ and $h_1 \in H_1$, using the Normality Test it is enough to prove that $h_1hh_1^{-1} \in H$. Using that $H/K \triangleleft H_1/K$ we obtain that

$$K(h_1hh_1^{-1}) = (Kh_1)(Kh)(Kh_1)^{-1} \in (Kh_1)(H/K)(Kh_1)^{-1} \subseteq H/K$$

Thus $K(h_1hh_1^{-1}) = Kh'$ for some $h' \in H$. So, $h_1hh_1^{-1} \in Kh' \subseteq H$ because $K \subseteq H$. Hence $H \triangleleft H_1$.

Conversely, let $H \triangleleft H_1$. Then $(Kh_1)(Kh)(Kh_1)^{-1} = K(h_1hh_1^{-1}) \in H/K$ for all $h_1 \in H_1$ and $h \in H$, because $h_1hh_1^{-1} \in H$. This shows that

$$(Kh_1)(H/K)(Kh_1)^{-1} \subseteq H/K \Rightarrow H/K \triangleleft H_1/K,$$

as desired.

(5) and (6) are tutorial problems.

□

Remark 1.23. Since the bijective correspondence Θ preserves inclusion, the lattice diagram of the subgroups of G/K has the same form as the lattice diagram of all subgroups of G containing K . In particular:

$$G \leftrightarrow G/K$$

$$K \leftrightarrow K/K = \{K\} - \text{the trivial subgroup of } G/K$$

Example 1.24. Let $C_{12} = \langle a \rangle$ be the cyclic group of order 12, and let $K = \langle a^6 \rangle$ and $K_1 = \langle a^4 \rangle$. **Draw the lattice diagram of all subgroups of C_{12} , and use the correspondence theorem to obtain the lattice diagram of all subgroups of C_{12}/K and C_{12}/K_1 .**

Solution: To determine the subgroups of C_{12} , we use the Fundamental Theorem of Finite Cyclic Groups. The subgroups of C_{12} are given by the positive divisors of 12, namely, 1, 2, 3, 4, 6, 12. The Fundamental Theorem of Finite Cyclic Groups allows us to conclude that C_{12} has exactly 6 subgroups. To determine the unique subgroup of each of these orders, remember that $\text{o}(a^d) = n/d$, provided $d|n$. We have that $\{1\} = \langle 1 \rangle = \langle a^{12} \rangle$, and

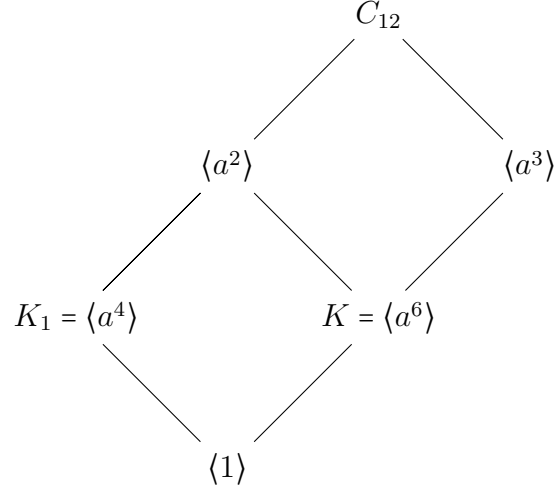
$$\text{Order 2: } 2 = \text{o}(a^d) = \frac{12}{d} \Rightarrow d = 6; \quad \text{the subgroup of order 2 is } \langle a^6 \rangle;$$

$$\text{Order 3: } 3 = \text{o}(a^d) = \frac{12}{d} \Rightarrow d = 4; \quad \text{the subgroup of order 3 is } \langle a^4 \rangle;$$

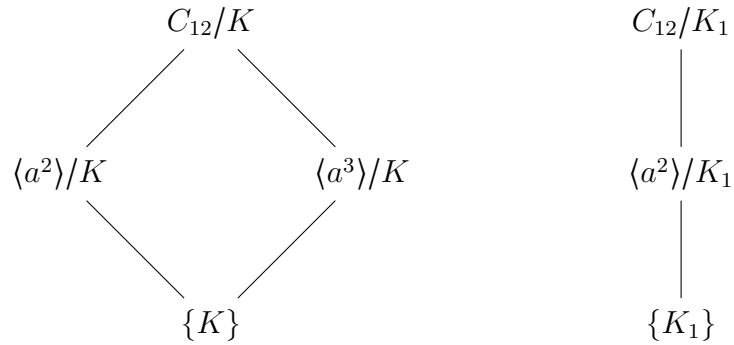
$$\text{Order 4: } 4 = \text{o}(a^d) = \frac{12}{d} \Rightarrow d = 3; \quad \text{the subgroup of order 4 is } \langle a^3 \rangle;$$

$$\text{Order 6: } 6 = \text{o}(a^d) = \frac{12}{d} \Rightarrow d = 2; \quad \text{the subgroup of order 6 is } \langle a^2 \rangle;$$

To draw the lattice diagram remember that $\langle a^m \rangle \subseteq \langle a^k \rangle \Leftrightarrow k|m$



To obtain the lattice diagram of C_{12}/K we just need to look at the subgroups H of C_{12} that contains K (upper left square) and consider H/K . Similarly, to picture the lattice diagram of C_{12}/K_1 , we focus our attention on the subgroups N of C_{12} containing K_1 , and we consider N/K_1 . Notice that in this case $N \in \{K_1, \langle a^2 \rangle, C_{12}\}$.

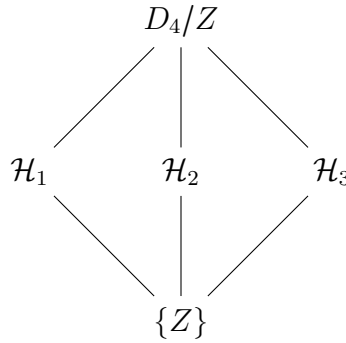


Example 1.25. Consider the octic group $D_4 = \{1, a, a^2, a^3, b, ba, ba^2, ba^3\}$, where $\text{o}(a) = 4$, $\text{o}(b) = 2$ and $aba = b$. For $Z = Z(D_4) = \{1, a^2\}$, **determine all the subgroups H of D_4 such that $Z \subseteq H \subseteq D_4$.**

Solution: From Example 1.21 we know that $D_4/Z = \{Z, Za, Zb, Z(ba)\} \cong K_4$, the Klein group. Hence, the subgroups of D_4/Z are

$$\{Z\}, \quad \mathcal{H}_1 = \{Z, Za\}, \quad \mathcal{H}_2 = \{Z, Zb\}, \quad \mathcal{H}_3 = \{Z, Z(ba)\}, \quad D_4/Z$$

The lattice diagram of D_4/Z is the following:



The Correspondence Theorem tells us that for $i \in \{1, 2, 3\}$ we can find a unique subgroup H_i of D_4 such that $Z \subseteq H_i$ and $\mathcal{H}_i = H_i/Z$. More precisely, from the Correspondence Theorem (1) we know that

$$H_1 = \{g \in D_4 \mid Zg \in \mathcal{H}_1\}, \quad H_2 = \{g \in D_4 \mid Zg \in \mathcal{H}_2\}, \quad H_3 = \{g \in D_4 \mid Zg \in \mathcal{H}_3\}$$

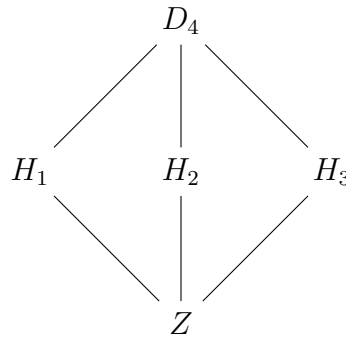
On the other hand from Example 1.9 we know that

$$Za^2 = Z1 = Z, \quad Za^3 = Za, \quad Z(ba^2) = Zb, \quad Z(ba^3) = Z(ba),$$

which yields that

$$H_1 = \{1, a, a^2, a^3\}, \quad H_2 = \{1, b, a^2, ba^2\}, \quad H_3 = \{1, ba, a^2, ba^3\}$$

The Correspondence Theorem asserts that (besides D_4 and Z) these are the only subgroups H of D_4 such that $Z \subseteq H \subseteq D_4$, and that the lattice diagram of these subgroups has exactly the same shape that the lattice diagram of D_4/Z . Moreover, the subgroups H_1 , H_2 and H_3 are normal in D_4 since $\mathcal{H}_1$, $\mathcal{H}_2$ and $\mathcal{H}_3$ are normal in D_4/Z .



Maximal normal subgroups

An important special case of the Correspondence Theorem describes when a factor group is simple. Recall that a group G is said to be **simple** if $G \neq \{1\}$ and its only normal subgroups are $\{1\}$ and G .

Let G be a group and $K \triangleleft G$. If $K \neq G$, then the factor group G/K is simple if and only if the only normal subgroups of G/K are the trivial subgroup $K/K = \{K\}$ and the whole group G/K . What does the Correspondence Theorem say in this case? Well ... It shows that the only normal subgroups H of G such that $K \subseteq H \subseteq G$ are $H = K$ and $H = G$. Such normal subgroups are called **maximal normal subgroups**. More precisely:

Definition 1.26. A normal subgroup $K \neq G$ is called a **maximal normal subgroup** if the only normal subgroups H of G satisfying that $K \subseteq H \subseteq G$ are $H = K$ and $H = G$.

Our previous discussion is the proof of the following theorem:

THEOREM 1.27. *A normal subgroup $K \triangleleft G$ is a maximal normal subgroup if and only if G/K is simple.*

Example 1.28. Find all the maximal normal subgroups of $\mathbb{Z}$.

Solution: Let us start by noticing that all the subgroups of $\mathbb{Z}$ are normal subgroups because $\mathbb{Z}$ is abelian. Moreover, we know that $\mathbb{Z}$ is a cyclic group so all its subgroups are cyclic, as well. It implies that all the subgroups of $\mathbb{Z}$ are of the form $n\mathbb{Z}$, where $n \in \mathbb{Z}$. Using Theorem 1.28 finding the maximal normal subgroups of $\mathbb{Z}$ is equivalent to determine the normal subgroups of $\mathbb{Z}$ such that the corresponding factor group is simple. In other words, we

need to determine when $\mathbb{Z}/n\mathbb{Z}$ is a simple group. From $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$ we obtain that the order of $\mathbb{Z}/n\mathbb{Z}$ is n . Now, from Lagrange's Theorem the order of any subgroup of $\mathbb{Z}/n\mathbb{Z}$ divides n . So, $\mathbb{Z}/n\mathbb{Z}$ will have the trivial subgroup $n\mathbb{Z}/n\mathbb{Z} = \{n\mathbb{Z}\}$ and the whole group $\mathbb{Z}/n\mathbb{Z}$ as normal subgroups if and only if $n = p$ is a prime. Thus, the maximal normal subgroups of $\mathbb{Z}$ are of the form $p\mathbb{Z}$ with p a prime.

1.8 The First Isomorphism Theorem

Let G and G' be groups. If there exists an epimorphism from G onto G' , we say that G' is a **homomorphic image** of G . Using this terminology, the canonical projection tells us that **any factor group G/H is a homomorphic image of G** . Put it this way, when we construct factor groups of G , we are, in fact, constructing homomorphic images of G .

Therefore, the factor group construction gives us a way of manufacturing homomorphic images of any group G . In this section, we will show that this is indeed a way of manufacturing all the homomorphic images of G . To be more precise, we will prove that every homomorphic image of G is isomorphic to a factor group of G .

As a consequence, G' is a homomorphic image of G if and only if G' is (or is isomorphic to) a factor group of G .

Let us start by proving a simple theorem.

THEOREM 1.29. *Let $\varphi : G \rightarrow G'$ be a group homomorphism with kernel K . Then $\varphi(a) = \varphi(b)$ if and only if $Ka = Kb$. In other words, any two elements a and b in G have the same image under φ if and only if they are in the same coset of K .*

PROOF. If a and b are two elements in G , then

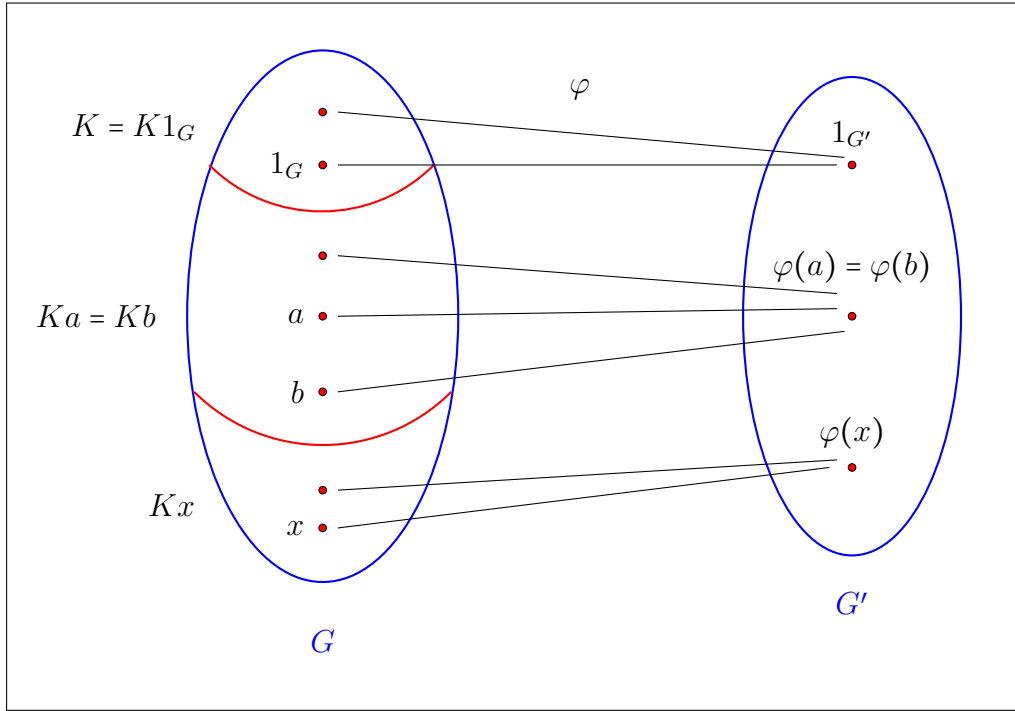
$$\varphi(a) = \varphi(b) \Leftrightarrow 1 = \varphi(a)\varphi(b)^{-1} = \varphi(a)\varphi(b^{-1}) = \varphi(ab^{-1}) \Leftrightarrow ab^{-1} \in K$$

Applying Cosets Property **6** we get

$$\varphi(a) = \varphi(b) \Leftrightarrow ab^{-1} \in K \Leftrightarrow Ka = Kb,$$

as desired. □

Given $\varphi : G \rightarrow G'$ an epimorphism with kernel K . Theorem 1.30 tells us that all the elements in a coset of K in G , say Kx have the same image under φ , and, conversely, elements which have the same image under φ are in the same coset of K .



We have then built a bijective mapping which matches cosets of K , namely, elements of the factor group G/K , with elements in G' . To be more precise, the coset Kx is matched with the element $\varphi(x) \in G'$. It remains to check that this correspondence is indeed an isomorphism.

THEOREM 1.30. FIRST ISOMORPHISM THEOREM.

Let $\varphi : G \rightarrow G'$ be an epimorphism. If K is the kernel of φ , then $G/K \cong G'$.

PROOF. To show that G/K is isomorphic to G' , we should find an isomorphism from G/K to G' . Let us try with the mapping $\tilde{\varphi} : G/K \rightarrow G'$ which matches each coset Kx with the element $\varphi(x)$. That is, $\tilde{\varphi}(Kx) = \varphi(x)$. We first need to make sure that $\tilde{\varphi}(Kx)$ is uniquely defined. To do so, we must make sure that if Ka is the same coset as Kb , then $\tilde{\varphi}(Ka) = \tilde{\varphi}(Kb)$. But, if $Ka = Kb$, then $\tilde{\varphi}(Ka) = \varphi(a) = \varphi(b) = \tilde{\varphi}(Kb)$ by Theorem 1.30. It proves that $\tilde{\varphi}$ is well-defined.

To finish, we show that $\tilde{\varphi}$ is an isomorphism.

- $\tilde{\varphi}$ is injective: If $\tilde{\varphi}(Ka) = \tilde{\varphi}(Kb)$, then $\varphi(a) = \varphi(b)$ and by Theorem 1.30, we get that $Ka = Kb$.

- $\tilde{\varphi}$ is surjective: every element of G' is of the form $\varphi(x) = \tilde{\varphi}(Kx)$.

Finally, $\tilde{\varphi}(Ka \cdot Kb) = \tilde{\varphi}(K(ab)) = \varphi(ab) = \varphi(a)\varphi(b) = \tilde{\varphi}(Ka)\tilde{\varphi}(Kb)$.

Therefore, $G/K \stackrel{\tilde{\varphi}}{\cong} G'$. □

The First Isomorphism Theorem asserts that **every homomorphic image of G is isomorphic to a quotient group of G** . If we are given an homomorphism $\varphi : G \rightarrow G'$, we can consider the restriction $\varphi : G \rightarrow \varphi(G)$ (which is an epimorphism) and apply the First Isomorphism Theorem to it. More precisely:

COROLLARY 1.31. Let $\varphi : G \rightarrow G'$ be a group homomorphism with kernel K . Then $G/K \cong \text{Im}\varphi$.

Let us see a few examples:

Example 1.32. Let $G = \left\{ \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} : a, b, c \in \mathbb{R} \right\}$. Then $Z(G) \cong \mathbb{R}$ and $G/Z(G) \cong \mathbb{R} \times \mathbb{R}$.

We first observe that G is a group under matrix multiplication. In fact:

- G is closed under matrix multiplication:

$$\begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & a' & b' \\ 0 & 1 & c' \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & a+a' & b'+ac'+b \\ 0 & 1 & c+c' \\ 0 & 0 & 1 \end{pmatrix}$$

- The identity matrix $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ is in G , and
- G is closed under inverses: $\begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & -a & ac-b \\ 0 & 1 & -c \\ 0 & 0 & 1 \end{pmatrix}$.

Since matrix multiplication is an associative operation, we can conclude that G is a group under matrix multiplication. Next, we compute the center of $Z(G)$. If $\begin{pmatrix} 1 & a' & b' \\ 0 & 1 & c' \\ 0 & 0 & 1 \end{pmatrix} \in Z(G)$, then

$$\begin{pmatrix} 1 & a' & b' \\ 0 & 1 & c' \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & a' & b' \\ 0 & 1 & c' \\ 0 & 0 & 1 \end{pmatrix}, \quad \forall a, b, c \in \mathbb{R}.$$

It implies that $b' + ac' + b = b + a'c + b'$, for all $a, b, c \in \mathbb{R}$. In particular, taking $a = 1$ and $c = 0$ gives $c' = 0$. Next, if we take $c = 1$ we obtain that $a' = 0$. Thus:

$$Z(G) = \left\{ \begin{pmatrix} 1 & 0 & b \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} : b \in \mathbb{R} \right\}$$

It is straightforward to check that the map $\varphi : \mathbb{R} \rightarrow Z(G)$ given by

$$b \mapsto \begin{pmatrix} 1 & 0 & b \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

is the desired isomorphism (fill in the details yourself!)

To finish, let us prove that $G/Z(G) \cong \mathbb{R} \times \mathbb{R}$. We will apply the Isomorphism Theorem. We then need to find an epimorphism from G to $\mathbb{R} \times \mathbb{R}$ with kernel $Z(G) \cong \mathbb{R}$. Taking into account the previous calculations, we consider the map $\psi : G \rightarrow \mathbb{R} \times \mathbb{R}$ given by

$$\psi \left(\begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \right) = (a, c).$$

Notice that ψ is surjective; if $(x, y) \in \mathbb{R} \times \mathbb{R}$, then $(x, y) = \psi \left(\begin{pmatrix} 1 & x & 0 \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix} \right) \in G$.

Next, we prove that ψ is a group homomorphism:

$$\begin{aligned} \psi \left[\begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & a' & b' \\ 0 & 1 & c' \\ 0 & 0 & 1 \end{pmatrix} \right] &= \psi \begin{pmatrix} 1 & a+a' & b'+ac'+b \\ 0 & 1 & c+c' \\ 0 & 0 & 1 \end{pmatrix} = \\ &= (a+a', c+c') = (a, c) + (a', c') = \psi \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} + \psi \begin{pmatrix} 1 & a' & b' \\ 0 & 1 & c' \\ 0 & 0 & 1 \end{pmatrix} \end{aligned}$$

Check that $\ker \psi = Z(G)$ (complete the details yourself). Thus, an application of the Isomorphism Theorem gives that $G/\ker \psi \cong \mathbb{R} \times \mathbb{R}$; $G/Z(G) \cong \mathbb{R} \times \mathbb{R}$.

1.9 The Second Isomorphism Theorem

Let G be a group and H, K subgroups of G . We define their **product** as

$$HK = \{hk \mid h \in H \text{ and } k \in K\}$$

Note that $H \subseteq HK$ and $K \subseteq HK$.

THEOREM 1.33. SECOND ISOMORPHISM THEOREM.

Let H and K be subgroups of a group G with $K \triangleleft G$. Then

- (i) HK is a subgroup of G .
- (ii) $K \triangleleft HK$
- (iii) $K \cap H \triangleleft H$ and the factor groups HK/K and $H/(H \cap K)$ are isomorphic.

$$\frac{HK}{K} \cong \frac{H}{H \cap K}$$

We first prove a preliminary lemma.

LEMMA 1.34. *Let H and K be subgroups of a group G . Then HK is a subgroup of G if and only if $KH \subseteq HK$.*

PROOF. Assume first that HK is a subgroup of G and take $kh \in KH$, where $k \in K$ and $h \in H$. Our goal is to show that $kh \in HK$. Applying that K and H are both subgroups of G , we have that $k^{-1} \in K$, $h^{-1} \in H$, which implies that

$h^{-1}k^{-1} \in HK$ and so $(h^{-1}k^{-1})^{-1} \in HK$ because we are assuming that HK is a subgroup of G . Thus:

$$kh = (h^{-1}k^{-1})^{-1} \in HK,$$

as desired. Conversely, suppose that $KH \subseteq HK$ and let us prove that HK is a subgroup of G . We will use the Subgroup Test. First notice that $1 = 1 \cdot 1 \in HK$. Next given $h_1k_1, h_2k_2 \in HK$ with $k_1, k_2 \in K$ and $h_1, h_2 \in H$ we have that

$$(h_1k_1)(h_2k_2) = h_1(k_1h_2)k_2, \quad (1.1)$$

but $k_1h_2 \in KH \subseteq HK$, and so $k_1h_2 = h'_2k'_1$ for some $k'_1 \in K$ and $h'_2 \in H$. Thus from (1.1) we get that

$$(h_1k_1)(h_2k_2) = h_1(k_1h_2)k_2 = h_1(h'_2k'_1)k_2 = (h_1h'_2)(k'_1k_2) \in HK.$$

To finish notice that

$$(h_1k_1)^{-1} = k_1^{-1}h_1^{-1} \in KH \subseteq HK.$$

The Subgroup Test allows us to conclude that HK is a subgroup of G . □

PROOF of the **SECOND ISOMORPHISM THEOREM**.

(i). In view of Lemma 3.16 the result will follow by showing that $KH \subseteq HK$. If $kh \in KH$, then $kh = h(h^{-1}kh) \in HK$ because $h^{-1}kh \in h^{-1}Kh = K$, since we are assuming that K is normal in G .

(ii). The subgroup K is normal in HK because K is normal in G (by hypothesis) and $K \subseteq HK$.

(iii). We are going to show that $K \cap H$ is a normal subgroup of H by viewing it as the kernel of a group homomorphism. Moreover, we are interested in finding an epimorphism from H to HK/K with kernel $H \cap K$ because (iii) will then follow by an application of the First Isomorphism Theorem.

We define $\varphi : H \rightarrow HK/K$ by $\varphi(h) = Kh$. Notice that $Kh \in HK/K$ because $h \in HK$. It is straightforward to check that φ is an homomorphism. Moreover, φ is an epimorphism because, if $x \in HK$, say $x = hk$ with $k \in K$ and $h \in H$. Then $Kx = K(hk) = (Kh)(Kk) = (Kh)(K) = Kh = \varphi(h)$. Let us now check that $\ker \varphi = K \cap H$. If $h \in \ker \varphi$, then $\varphi(h) = K$, that is, $Kh = K$, which

by Cosets Property **7** is equivalent to say that $h \in K$. Thus $h \in H \cap K$ and $\ker \varphi \subseteq H \cap K$. The other containment holds by Cosets Property **7**. Thus $\ker \varphi = H \cap K$, which tells us that $H \cap K \triangleleft H$. By the First Isomorphism Theorem we obtain that

$$\frac{H}{H \cap K} \cong \frac{HK}{K},$$

as desired. $\square$

We close the section with an application of the Second Isomorphism Theorem. A group G is called a **metacyclic group** if there exists a normal subgroup K of G such that both K and G/K are cyclic. Every cyclic group is metacyclic (take $K = \{1\}$). We can use the Second Isomorphism Theorem to show that **every subgroup of a metacyclic group is again metacyclic**.

Let G be a metacyclic group and H a subgroup of G . We know that there exists $K \triangleleft G$ such that K and G/K are both metacyclic. Applying the Second Isomorphism Theorem to G , $K \triangleleft G$ and H we get that HK is a subgroup of G having K as a normal subgroup, $H \cap K \triangleleft H$ and the factor groups $H/(H \cap K)$, HK/K are isomorphic.

We know that K is a cyclic group and $H \cap K$ is a subgroup of it, so $H \cap K$ is cyclic. On the other hand, the factor group HK/K is a subgroup of G/K , which is cyclic and so HK/K is cyclic. The isomorphism $H/(H \cap K) \cong HK/K$ implies that $H/(H \cap K)$ is cyclic. Notice that we are done, H is metacyclic since its subgroup $H \cap K$ is cyclic and the factor group $H/(H \cap K)$ is cyclic.

1.10 The Third Isomorphism Theorem

THEOREM 1.35. THIRD ISOMORPHISM THEOREM.

Let $K \subseteq H \subseteq G$ be groups, where $K \triangleleft G$ and $H \triangleleft G$. Then $H/K \triangleleft G/K$ and

$$\frac{G/K}{H/K} \cong G/H$$

PROOF. We define $\varphi : G/K \rightarrow G/H$ by $\varphi(Kg) = Hg$ for all $g \in G$. The first thing that we need to do is to check that φ is well defined, that is, it does not depend on the representative of the coset. If $Kg = Kg_1$, we then need

to show that $Hg = Hg_1$. From $Kg = Kg_1$ we can conclude that $gg_1^{-1} \in K$ but $K \subseteq H$, so $gg_1^{-1} \in H$, which implies that $Hg = Hg_1$.

It is straightforward to show that φ is an epimorphism (complete the details). Next, we claim that $\ker \varphi = H/K$. In fact, by definition

$$\ker \varphi = \{Kg \in G/K \mid \varphi(Kg) = H\}$$

But $\varphi(Kg) = Hg = H$ if and only if $g \in H$, and so $\ker \varphi = H/K$. By the First Isomorphism Theorem we get that

$$\frac{G/K}{\ker \varphi} \cong \text{Im} \varphi, \quad \text{that is,} \quad \frac{G/K}{H/K} \cong G/H.$$

□

As an application of the Third Isomorphism Theorem, we will show that every factor group of a metacyclic groups is also metacyclic. Assume that G is a metacyclic group and let $K \triangleleft G$ such that K and G/K are both cyclic groups. Consider G/N with $N \triangleleft G$. The Second Isomorphism Theorem tells us that $KN \leq G$ and $KN/N \cong K/(N \cap K)$. In this case, because both K and N are normal in G , we have that KN is also normal in G . In fact, for $g \in G$ we have that

$$(KN)g = K(Ng) = K(gN) = (Kg)N = (gK)N = g(KN),$$

which shows that $KN \triangleleft G$. Now, from the Correspondence Theorem we obtain that $KN/N \triangleleft G/N$. Since K is cyclic, the factor group $K/(N \cap K)$ is cyclic and so is KN/N . On the other hand, by the Third Isomorphism Theorem

$$\frac{G/N}{KN/N} \cong G/KN,$$

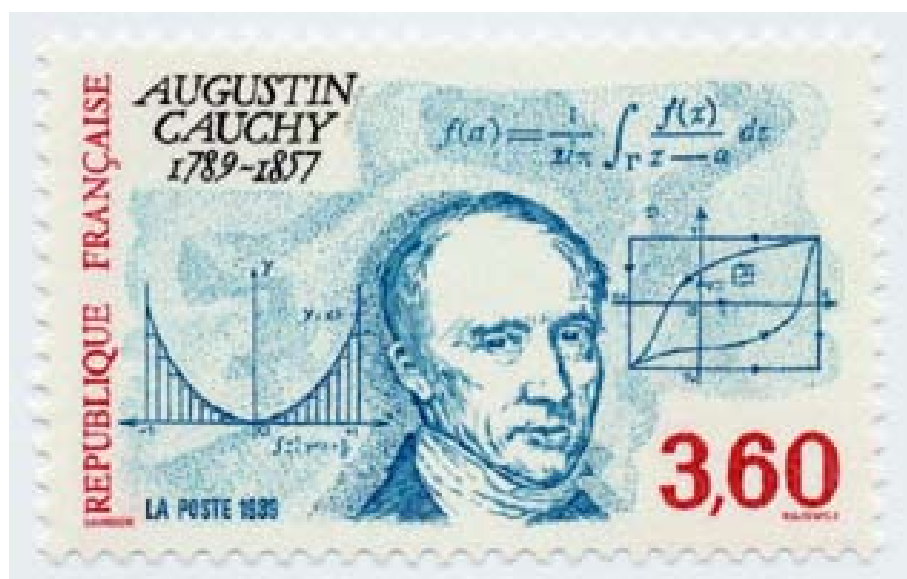
which is a cyclic group since

$$G/KN \cong \frac{G/K}{KN/K}.$$

Notice that we have already showed that G/N is metacyclic because its subgroup KN/N is cyclic and the factor group $\frac{G/N}{KN/N}$ is also cyclic.

Chapter II

Cauchy's Theorem



Chapter 2

Cauchy's Theorem

Let p be a prime and G a group of order p^n . A quick application of Lagrange's Theorem reveals that the order of any element of G is a power of p . The converse, known as *Cauchy's Theorem*, is also true. If every element of a finite group G has order a power of p , then $|G| = p^n$ for some $n \geq 0$.

To prove Cauchy's Theorem we need to develop some machinery and learn a bit more about groups of p -power order.

2.1 Group Action on a Set

We have already seen examples of how groups can act on things like, for example, the group of symmetries of a square. In this section, we introduce the general notion of a group acting on a set and present some examples.

Last year we defined a (closed) binary operation $*$ on a set S to be a map from $S \times S$ into S . The map $*$ provides us a rule for “multiplying” an element s_1 in S and an element s_2 in S to obtain a new element $s_1 * s_2$ in S . In a more general way, for any sets A , B and C we can view a map $*$: $A \times B \rightarrow C$ as defining a “multiplication”, where any element $a \in A$ times any element $b \in B$ gives rise to some element $c \in C$, usually denoted by $a * b = c$, or simply $ab = c$. Here we are interested in the case of having a group G , a set X , a map $*$: $G \times X \rightarrow X$. We will write $*(g, x)$ as $g * x$, or gx .

Definition 2.1. Let X be a set and G a group. An **action of G on X** is a map $*$: $G \times X \rightarrow X$ (written as gx for all $g \in G$ and $x \in X$) such that

1. $(g_1 g_2) * x = g_1 * (g_2 * x)$ for all $g_1, g_2 \in G$ and $x \in X$.

2. $1 * x = x$ for all $x \in X$.

Under these conditions we say that X is a G -set.

Examples 2.2.

1. If X is any set and G is any group, the **trivial action** is defined by $g * x = x$ for all $x \in X$ and $g \in G$.
2. Let G be a group and let $X = G$. The operation in G gives a group action of G on itself, where the action on g_2 by g_1 is given by left multiplication. To be more precise, $*(g_1, g_2) = g_1 g_2$ (or $g_1 + g_2$ if G is an additive group).
3. Let V a vector space over a field F . The axioms $(\lambda\mu)v = \lambda(\mu v)$ and $1v = v$ for scalars $\lambda, \mu \in F$, and a vector v show that V is an $F^\times$ -set, where $F^\times$ denotes the multiplicative group of nonzero scalars.
4. Let X be any set and H a subgroup of the group S_X of all permutations of X . Then X is an H -set, where the action of $\sigma \in H$ on X is its action as an element of S_X , that is, $\sigma * x = \sigma(x)$ for all $x \in X$. In particular, $\{1, 2, \dots, n\}$ is an S_n -set.

Our first result shows that for every G -set X , the map $\sigma_g : X \rightarrow X$ defined by $\sigma_g(x) = g * x$ is a permutation of X , and that there is a homomorphism from G to S_X .

THEOREM 2.3. *Let G be a group acting on a set X .*

- (i) *For each $g \in G$, the map $\sigma_g : X \rightarrow X$ defined by $\sigma_g(x) = g * x$ for $x \in X$ is a permutation of X .*
- (ii) *The map $\phi : G \rightarrow S_X$ given by $\phi(g) = \sigma_g$ is a homomorphism.*

PROOF. (i) Given $g \in G$, to show that σ_g is a permutation of X , we just need to prove that σ_g is bijective. But you can easily check that $\sigma_g^{-1} = \sigma_{g^{-1}}$.

(ii) We must show that $\phi(g_1 g_2) = \phi(g_1) \circ \phi(g_2)$ for all g_1, g_2 . To do so, we need to prove that both permutations act in the same way, that is, $\phi(g_1 g_2)(x) =$

$(\phi(g_1) \circ \phi(g_2))(x)$. Take $x \in X$ and apply the two conditions in Definition 2.1 to obtain

$$\begin{aligned}\phi(g_1 g_2)(x) &= \sigma_{g_1 g_2}(x) = (g_1 g_2) * x = g_1 * (g_2 * x) = g_1 * \sigma_{g_2}(x) = \sigma_{g_1}(\sigma_{g_2}(x)) = \\ &= (\sigma_{g_1} \circ \sigma_{g_2})(x) = (\phi(g_1) \circ \phi(g_2))(x),\end{aligned}$$

as desired. $\square$

Remark 2.4. If X is a G -set, then **the elements of G fixing every element of X is a normal subgroup of G** , since it coincides with the kernel of the homomorphism $\phi : G \rightarrow S_X$. In fact,

$$\ker(\phi) = \{g \in G \mid \sigma_g = \text{Id}_X\} = \{g \in G \mid g * x = \sigma_g(x) = x \text{ for all } x \in X\}$$

We call $\ker(\phi)$ the **kernel of the action**. If $\ker(\phi) = \{1\}$, then the identity element of G is the only element of G that leaves every $x \in X$ fixed; in such a case we say that G **acts faithfully** on X .

Definition 2.5. Let G be a group acting on a set X . For each $x \in X$ the **stabilizer of x in G** , denoted $S(x)$, is the set of elements of G that fix the element x , that is, $S(x) = \{g \in G \mid g * x = x\}$. It is easy to check that $S(x)$ is a subgroup of G .

We next prove that the action of a group G on a set X gives rise to an equivalence relation on X .

THEOREM 2.6. *Let G be a group acting on a set X . The relation on X*

$$x_1 \sim x_2 \text{ if and only if } x_2 = g * x_1 \text{ for some } g \in G$$

is an equivalence relation on X . For each $x \in X$, the number of elements in the equivalence class Gx of x is $|G : S(x)|$, the index of the stabilizer of x .

PROOF. Condition 2 in Definition 2.1 shows that $\sim$ is reflexive. If $x_1 \sim x_2$, then $x_2 = g * x_1$ for some $g \in G$. Then $g^{-1} * x_2 = g^{-1} * (g * x_1) = (g^{-1}g) * x_1 = 1 * x_1 = x_1$, and $\sim$ is symmetric. Finally if $x_1 \sim x_2$ and $x_2 \sim x_3$, then $x_2 = g * x_1$ and $x_3 = g' * x_2$ for some $g, g' \in G$. Thus $x_3 = g' * x_2 = g' * (g * x_1) = (g'g) * x_1$, so $x_1 \sim x_3$ and $\sim$ is transitive.

Recall that the equivalence class $G * x$ of x is just the subset of X consisting of all the elements related to x via $\sim$, that is,

$$G * x = \{x' \in X \mid x \sim x'\} = \{g * x \mid g \in G\}.$$

We are going to define a bijection ψ between the left cosets of $S(x)$ in G and the elements of $G * x$. Let $x_1 \in G * x$, then $x_1 = g_1 * x$ for some $g_1 \in G$. We define $\psi(x_1)$ to be the left coset of $S(x)$ of $S(x)$, i.e., $\psi(x_1) = g_1 S(x)$.

To show that ψ is injective, assume that $\psi(x_1) = \psi(x_2)$ for some $x_1, x_2 \in [x]$. Then there exists $g_1, g_2 \in G$ such that $x_1 = g_1 * x$ and $x_2 = g_2 * x$. From $\psi(x_1) = \psi(x_2)$ we derive $g_1 S(x) = g_2 S(x)$. In particular, $g_2 \in g_1 S(x)$, and so $g_2 = g_1 g$ for some $g \in S(x)$. Thus $x_2 = g_2 * x = (g_1 g) * x = g_1 * (g * x) = g_1 * x = x_1$, and ψ is injective.

Finally, we show that each left coset of $S(x)$ in G is of the form $\psi(x_1)$ for some $x_1 \in G * x$. Let $g_1 S(x)$ be a left coset. Then $x_1 = g_1 * x \in G * x$ and satisfies that $\psi(x_1) = g_1 S(x)$. $\square$

Definition 2.7. Let X be a G -set. The equivalence class $G * x = \{g * x \mid g \in G\}$ of x under the equivalence relation described in the previous theorem is called the **orbit of x under G** .

The action of G on X is called **transitive** if there is only one orbit, that is, for all $x_1, x_2 \in X$ there exists $g \in G$ such that $g * x_1 = x_2$.

Let X be a G -set. Then from Theorem 2.6 we can characterize the elements x of X such that its orbit Gx is a singleton. In fact:

$$G * x = \{x\} \Leftrightarrow |G : S(X)| = 1 \Leftrightarrow S(x) = G \Leftrightarrow g * x = x \text{ for all } g \in G.$$

The set of all such elements x is denoted

$$X_f = \{x \in X \mid g * x = x \text{ for all } g \in G\}.$$

and is called the **fixed subset** of X under the action of G . Notice that X_f is the union of the singleton orbits.

Examples 2.8.

1. If a group G acts trivially on a set X , then $S(x) = G$ for all $x \in X$, and the orbits are precisely the elements of X .
2. The symmetric group S_n acts transitively in its usual action as permutations on $X_n = \{1, 2, \dots, n\}$.

THEOREM 2.9. ORBIT DECOMPOSITION THEOREM.

Let G be a group acting on a finite set $X \neq \emptyset$. Let $G * x_1, G * x_2, \dots, G * x_n$ denote the nonsingleton orbits. Then

$$|X| = |X_f| + \sum_{i=1}^n |G : S(x_i)|$$

PROOF. The fixed subset is the union of the singleton orbits. Because the orbits partition X , we have that $|X| = |X_f| + \sum_{i=1}^n |G * x_i|$. But from Theorem 2.6 we have that $|G * x_i| = |G : S(x_i)|$ for each i . Thus $|X| = |X_f| + \sum_{i=1}^n |G : S(x_i)|$, as required. $\square$

2.2 Conjugation

Let G be a group. Then G can act on itself (i.e. $X = G$) by conjugation:

$$g * x = gxg^{-1} \text{ for all } g, x \in G,$$

where gxg^{-1} is computed in the group G , as usual. This formula satisfies the two conditions in Definition 2.1. In fact, clearly $1 * g = g$ for all $g \in G$, and $g_1 * (g_2 * x) = g_1 * (g_2 x g_2^{-1}) = g_1 (g_2 x g_2^{-1}) g_1^{-1} = (g_1 g_2) x (g_2^{-1} g_1^{-1}) = (g_1 g_2) x (g_1 g_2)^{-1}$, for all $g_1, g_2, x \in G$.

Definition 2.10. Let G be a group and a, b two elements in G . We say that a and b are **conjugate in G** if $b = gag^{-1}$ for some $g \in G$. That is to say that a and b are in the same orbit of G acting on itself by conjugation.

Elements of the form gag^{-1} are called **conjugates** of a . Roughly speaking a conjugate of a is any product consisting of a sandwiched between any element and its inverse.

The orbits of G acting on itself by conjugation are called the **conjugacy classes** of G . The conjugacy class of $a \in G$ is denoted by $\text{class } a$. Notice that it consists on the conjugates of a :

$$\text{class } a = G * a = \{g * a \mid g \in G\} = \{gag^{-1} \mid g \in G\}$$

The conjugacy classes constitute a partition of G .

Examples 2.11.

1. If G is an abelian group, then the action of G on itself by conjugation is the trivial action, and $\text{class } a = \{a\}$ for each $a \in G$.
2. $\text{class } 1 = \{1\}$ in any group. More generally,

$$\text{class } a = \{a\} \Leftrightarrow a \in Z(G)$$

2. Let $G = S_n$. Then two permutations in S_n are conjugate if and only if they have the same cycle structure; that is, when factored into disjoint cycles they have the same number of cycles of each length.

It follows from the fact that the conjugate of an r -cycle of length is an r -cycle, as well. More precisely, if $(k_1 k_2 \dots k_r)$ is an r -cycle and $\sigma \in S_n$, then $\sigma(k_1 k_2 \dots k_r)\sigma^{-1} = (\sigma k_1 \sigma k_2 \dots \sigma k_r)$. If $\alpha = \gamma_1 \dots \gamma_s$ is the decomposition of α into disjoint cycles, then

$$\sigma \alpha \sigma^{-1} = (\sigma \gamma_1 \sigma^{-1})(\sigma \gamma_2 \sigma^{-1}) \dots (\sigma \gamma_s \sigma^{-1})$$

It remains to check that $(\sigma \gamma_i \sigma^{-1})$ and $(\sigma \gamma_j \sigma^{-1})$ are disjoint. This is left as an exercise.

In particular, the conjugacy classes of S_3 are

$$\begin{aligned} \text{class } \varepsilon &= \{\varepsilon\}, \\ \text{class } (1 \ 2) &= \{(1 \ 2), (1 \ 3), (2 \ 3)\}, \\ \text{class } (1 \ 2 \ 3) &= \{(1 \ 2 \ 3), (1 \ 3 \ 2)\}. \end{aligned}$$

THEOREM 2.12. Let G be a group and $g \in G$. The conjugation mapping $\lambda_g : G \rightarrow G$ defined by $\lambda_g(x) = gxg^{-1}$ is an automorphism of G .

PROOF. If x, y are in G , then

$$\lambda_g(xy) = g(xy)g^{-1} = gx(g^{-1}g)yg^{-1} = (gxg^{-1})(gyg^{-1}) = \lambda_g(x)\lambda_g(y),$$

which shows that λ_g is an homomorphism. Moreover, λ_g is bijective because $\lambda_g^{-1} = \lambda_{g^{-1}}$. $\square$

Such conjugation automorphisms are called **inner automorphisms**. The set consisting of all inner automorphisms of G , namely, $\text{Inn}(G) = \{\lambda_g \mid g \in G\}$ is a normal subgroup of the group of automorphisms of G . In terms of inner automorphisms the conjugacy class of $a \in G$ can be written as

$$\text{class } a = \{\lambda_g(a) \mid g \in G\}.$$

COROLLARY 2.13. *Let G be a group and $a, b \in G$. If a and b are conjugate, then $\text{o}(a) = \text{o}(b)$. Moreover, all the elements in a conjugacy class have the same order.*

PROOF. If a and b are conjugate, then $b = gag^{-1} = \lambda_g(a)$. The result now follows from Theorem 2.12 because automorphisms preserve the order of the elements. $\square$

Example 2.14. Find the **conjugacy classes** of $D_3 = \{1, a, a^2, b, ba, ba^2\}$, where $\text{o}(a) = 3$, $\text{o}(b) = 2$ and $aba = b$.

Solution: Recall that $a^k b = ba^{3-k}$ for all $k \in \mathbb{Z}$. As a and a^2 are the only elements with order 3, we have that $\text{class } a \subseteq \{a, a^2\}$. But from $a^2 b = ba$ we obtain that $a^2 = bab^{-1} \in \text{class } a$ and so $\text{class } a = \{a, a^2\}$. The elements ba and ba^2 have both order 2, so $\text{class } b \subseteq \{b, ba, ba^2\}$. On the other hand, from $aba = b$ we obtain that $aba^2 = ba$ and since $a^2 = a^{-1}$, we obtain that $ba \in \text{class } b$. To finish, from $a^2 b = ba$ we derive that $a^2 ba^{-2} = ba^{-1} = ba^2$ and $ba^2 \in \text{class } b$. Thus: $\text{class } b = \{b, ba, ba^2\}$. The conjugacy classes of D_3 are

$$\text{class } 1 = \{1\},$$

$$\text{class } a = \{a, a^2\},$$

$$\text{class } b = \{b, ba, ba^2\}.$$

We can generalize the conjugacy action by allowing G acts on the set $\mathcal{P}(G)$ of all subsets of G . If T is any subset of G and $g \in G$, we define $g * T = gTg^{-1}$, where $gTg^{-1} = \{gtg^{-1} \mid t \in T\}$.

Definition 2.15. We say that two subsets T and U of G are **conjugate in G** if $U = gTg^{-1}$ for some $g \in G$. The subsets of the form gTg^{-1} are called **conjugates of T** . As before, the conjugation relation is an equivalence relation.

Normal subgroups are self-conjugate: if $K \triangleleft G$, then $gKg^{-1} = K$ for all $g \in G$. In particular, it says that K contains the conjugacy class of each of its elements. In fact, if $a \in K$ then

$$\text{class } a = \{gag^{-1} \mid g \in G\} \subseteq K$$

The converse is also true.

THEOREM 2.16. *Let H be a subgroup of a group G . Then $H \triangleleft G$ if and only if H is a union of conjugacy classes.*

PROOF. If H is normal, then H is the union of the conjugacy classes of its elements. Conversely, assume that H is a union of conjugacy classes. Suppose that $H = \text{class } a_1 \cup \dots \cup \text{class } a_n$. We use the Normality Test: for $g \in G$ we need to show that $gHg^{-1} \subseteq H$. If $h \in H$, let $h \in \text{class } a_i$ (remember that the conjugacy classes are disjoint sets), then $h = ba_i b^{-1}$ for some $b \in G$. Thus:

$$ghg^{-1} = g(ba_i b^{-1})g^{-1} = (gb)a_i(b^{-1}g^{-1}) = (gb)a_i(gb)^{-1} \in \text{class } a_i \subseteq H,$$

as desired. □

Theorem 2.16 provides us a method to determine the normal subgroups of a group. For example, by Theorem 2.16 any normal subgroup K of D_3 must be an union of conjugacy classes of D_3 , which are $\{1\}$, $\{a, a^2\}$, $\{b, ba, ba^2\}$. On the other hand, the order of K must divide 6 (the order of D_3) by Lagrange's Theorem. So, the normal subgroups of D_3 are $\{1\}$, $\{1, a, a^2\}$ and D_3 .

Let G be a group and T a subset of G . Theorem 2.6 tells us that the number of conjugates of T equals the index $|G : S(T)|$ of the stabilizer $S(T)$

of T . In our case, for action by conjugation the stabilizer

$$S(T) = \{g \in G \mid gTg^{-1} = T\} = N_G(T)$$

is called the **normalizer of T in G** . For the particular case of conjugation by elements of G , Theorem 2.6 yields that the number of conjugates of an element a of G is the index of the $|G : S(a)|$ of the stabilizer $S(a) = S(\{a\})$ of a , which is called the **centralizer of a in G** and denoted $C_G(a)$ or $C(a)$.

$$S(a) = S(\{a\}) = \{g \in G \mid gag^{-1} = a\} = N_G(\{a\}) = C_G(a) = C(a), \\ |G : C(a)| = |\text{class } a|$$

Notice that $C(a)$ consists on the elements of G that commutes with a (saying $gag^{-1} = a$ is equivalent to say that $ga = ag$). Moreover we have that

$$\text{class } a = \{a\} \Leftrightarrow a \in Z(G) \Leftrightarrow C(a) = G$$

For acting by conjugation the **fixed subset of $G(= X)$ becomes the center of G** . In fact:

$$G_f = \{a \in G \mid gag^{-1} = a \text{ for all } g \in G\} = \\ = \{a \in G \mid ga = ag \text{ for all } g \in G\} = Z(G)$$

Let G be a finite group and $\text{class } a_1, \text{class } a_2, \dots, \text{class } a_n$ be the nonsingleton conjugacy classes. The Orbit Decomposition Theorem yields

$$|G| = |Z(G)| + \sum_{i=1}^n |G : C(a_i)| \quad (\text{CE})$$

Equation (CE) is called the **Class Equation**. Note that all the summands on the right hand side of the Class Equation, namely $|G : C(a_i)|$, are divisors of $|G|$ because they are indices of subgroups of G .

Examples 2.17.

1. The Class Equation gives no information in an abelian group since conjugation is the trivial action and all conjugacy classes have size 1.
2. Let us consider the group D_3 . We computed the conjugacy classes in Example 2.14. Recall that $[1] = \{1\}$, $[a] = \{a, a^2\}$, and $[b] = \{b, ba, ba^2\}$. The Class Equation tells us that the cardinal of D_3 is $1 + 2 + 3 = 6$, as expected. The center is trivial in this case.

2.3 Cauchy's Theorem

We are now in a position to prove Cauchy's Theorem. In some texts, Cauchy's Theorem is first proved for abelian groups and using this abelian version they then provide a proof for the general case. Here, we will use group actions to give a proof that works for all the groups abelian or not. This beautiful proof of Cauchy's Theorem is due to J. H. McKay. We first need a preliminary lemma.

LEMMA 2.18. *Let G be a group of order p^n and X be a finite G -set. Then $|X| \equiv |X_f| \pmod{p}$.*

PROOF. The Orbit Decomposition Theorem tells us that

$$|X| = |X_f| + \sum_{i=1}^m |G : S(x_i)|,$$

which implies that $|X| - |X_f| = \sum_{i=1}^m |G : S(x_i)|$. The result now follows because p divides $|G : S(x_i)|$, since they are indices of subgroups of G . $\square$

THEOREM 2.19. CAUCHY'S THEOREM

Let G be a finite group. If a prime p divides the order of G , then G has an element of order p and, consequently, a subgroup of order p .

PROOF. Let X be the set of all p -tuples $(g_1, \dots, g_p)$ of elements of G such that the product of the coordinates in G equals 1 (the identity element of G). That is,

$$X = \{(g_1, \dots, g_p) \mid g_i \in G \text{ and } g_1 g_2 \dots g_p = 1\}$$

We claim that p divides $|X|$. In fact, $|X| = |G|^{p-1}$ because $g_1, \dots, g_{p-1}$ can be arbitrary elements of G but once those are fixed, g_p has necessarily to be $(g_1 g_2 \dots g_{p-1})^{-1}$. Now since p divides $|G|$, then p divides $|G|^{p-1} = |X|$.

Let $\sigma = (1 \ 2 \dots p) \in S_p$ and let σ acts on X by

$$\sigma(g_1, \dots, g_p) = (g_{\sigma(1)}, \dots, g_{\sigma(p)}) = (g_2, g_3, \dots, g_p, g_1)$$

The element $(g_2, g_3, \dots, g_p, g_1)$ is in X . In fact, from $g_1 g_2 \dots g_p = 1$ we get that $g_1^{-1} = g_2 \dots g_p$, and so $g_2 g_3 \dots g_p g_1 = 1$. Then we can define the action of the subgroup $\langle \sigma \rangle$ of S_p on X just by iterating the action of σ on X in the natural

way. Let us now consider the fixed subset X_f of X under the action of $\langle \sigma \rangle$. By Lemma 2.18 we have that $|X| \equiv |X_f| \pmod{p}$. Now since p divides $|X|$ we can then conclude that p also divides $|X_f|$. On the other hand a p -tuple $(g_1, \dots, g_p)$ belongs to X_f if and only if it is fixed by σ , and this happens if and only if $g_1 = g_2 = \dots = g_p$. To finish, since p divides $|X_f|$, the set X_f must contain at least p elements. So, there exists $a \in G$ with $a \neq 1$ such that $(a, \dots, a) \in X_f$. Thus $a^p = 1$, so a has order p , and the subgroup $\langle a \rangle$ of G has order p . $\square$

2.4 p -Groups

Cauchy's Theorem can be used to characterize groups having order the power of a prime. This might be one of the most important of its applications.

Definition 2.20. Let p be a prime. A group G is called a **p -group** if every element of G has order a power of the prime p . A subgroup of a group G is a **p -subgroup of G** if the subgroup is itself a p -group.

Cauchy's Theorem provides this characterization of finite p -groups.

COROLLARY 2.21. *Let G be a finite group and p a prime. Then G is a p -group if and only if the order of G is a power of p .*

PROOF. Assume that G is a p -group. Then $\text{o}(g)$ is a power of p for all $g \in G$. We claim that $|G|$ is a power of p . Suppose on the contrary that $|G|$ is not a power of p . Then we can find a prime q different from p that divides $|G|$. Applying Cauchy's Theorem we can guarantee the existence of an element of G with order q , but this contradicts our hypothesis. Hence, $|G|$ is a power of p . Conversely, if $|G| = p^n$, then an application of Lagrange's Theorem reveals that every element of G has order a power of p . $\square$

We next prove that p -groups have nontrivial centers.

THEOREM 2.22. *If p is a prime and $G \neq \{1\}$ is a finite p -group. Then $Z(G) \neq \{1\}$.*

PROOF. By Corollary 2.21 we have that G has order a power of p . In particular, p divides $|G|$. On the other hand, $|G| \equiv |Z(G)| \pmod{p}$ by Lemma 2.18. Thus p divides $|Z(G)|$, and so $Z(G) \neq \{1\}$. $\square$

Last year we proved that (up to isomorphisms) there are only two groups of order 4: the cyclic group C_4 and the Klein group $K_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \cong C_2 \times C_2$. Using Theorem 2.22 we can generalize this result.

THEOREM 2.23. *Let G be a group with $|G| = p^2$, where p is a prime. Then G is abelian and either $G \cong \mathbb{Z}_{p^2}$ or $G \cong \mathbb{Z}_p \times \mathbb{Z}_p$.*

PROOF. To prove that G is abelian, we will show that $Z(G) = G$. From Theorem 2.22 we know that $Z(G) \neq \{1\}$. By Lagrange's Theorem either $|Z(G)| = p$ or $|Z(G)| = p^2$. If $|Z(G)| = p$, then $Z(G) \neq G$. On the other hand, we have that $G/Z(G)$ is a cyclic group (it has order p), which implies that G is abelian (by the properties of factor groups). But this is impossible because $Z(G) \neq G$. Hence $|Z(G)| = p^2$, which implies that $G = Z(G)$, and G is abelian.

If G has an element of order p^2 , then G is cyclic, and so $G \cong \mathbb{Z}_{p^2}$. Otherwise, $\text{o}(g) = p$ for all $g \in G$. Then $g^p = 1$ for all $g \in G$. Choose $a \in G$ with $a \neq 1$ and write $\langle a \rangle \cong \mathbb{Z}_p$; now choose $b \in G$ with $b \notin \langle a \rangle$. Since $|\langle a, b \rangle| > |\langle a \rangle| = p$, Lagrange's Theorem yields that $|\langle a, b \rangle| = p^2$, that is, $G = \langle a, b \rangle$. Lastly, the map $(a^n, b^m) \mapsto a^n b^m$ provides an isomorphism from $\langle a \rangle \times \langle b \rangle \cong \mathbb{Z}_p \times \mathbb{Z}_p$ onto $\langle a, b \rangle = G$. This finishes the proof. $\square$

Notice that Theorem 2.23 is not true for groups of order p^3 . For instance, for $p = 2$ the nonabelian group D_4 has order 2^3 .

Chapter III

Sylow Theorems



Chapter 3

Sylow Theorems

Lagrange's Theorem asserts that the order of each subgroup of a finite group G is a divisor of $|G|$, the order of G . The converse of Lagrange's Theorem does not hold in general. If G is a finite group and d is a divisor of $|G|$, then there might not be a subgroup of G with order d . The simplest example when this situation happens is the alternating group A_4 of order $|A_4| = 12$, which has no subgroup of order 6. The Sylow Theorems give a weak converse. They show that if a divisor d of $|G|$ is a power of a prime, then G contains a subgroup of order d . (Notice that 6 is not a power of a prime.) In addition, the Sylow Theorems provide some information about the number of such subgroups and their relationship to each other.

3.1 The Sylow Theorems

The Norwegian mathematician Peter Ludwig Sylow (1832-1918) discovered in 1872 that a “converse result” to Lagrange's Theorem is true when d is a prime power. More precisely, if p is a prime number such that p^k divides the order of a group G , then G must contain a subgroup of order p^k . Sylow stated the theorems in terms of permutation groups (since the abstract definition of a group had not been given yet). Later on, in 1887, Georg Frobenius re-proved the Sylow theorems for abstract groups. His proofs made use of the Class Equation. The proof presented here is a more modern proof, which uses the theory of group actions.

Let G be a group and $\mathcal{S}(G)$ the **collection of all subgroups of G** . Recall from the previous chapter that we can let G act on $\mathcal{S}$ by conjugation, that is, if $H \in \mathcal{S}(G)$ and $g \in G$, then g acting on H yields the conjugate subgroup gHg^{-1} of H . To avoid confusion with the notation of left cosets, we will not write this action as gH . Along this chapter we will use left cosets because they are more natural when dealing with group actions.

We called the stabilizer of $H \in \mathcal{S}(G)$ the **normalizer of H in G** .

$$N(H) = \{g \in G \mid gHg^{-1} = H\}$$

Recall that $N(H)$ is a subgroup of G and notice that $H \triangleleft N(H)$. Actually, the normalizer receives its name because it is indeed the **largest subgroup of G having H as a normal subgroup**. Notice that $N(H) = G$ if and only if $H \triangleleft G$.

LEMMA 3.1. *Let H be a subgroup of a group G . Then $g \in N(H)$ if and only if $ghg^{-1} \in H$ for all $h \in H$.*

PROOF. Let $g \in G$ such that $ghg^{-1} \in H$ for all $h \in H$. Then the inner automorphism $\lambda_g(x) = gxg^{-1}$ of G restricted to H gives rise to an automorphism of H . Thus $H = \lambda_g(H) = gHg^{-1}$ and $g \in N(H)$. The converse trivially holds. $\square$

LEMMA 3.2. *Let H be a p -subgroup of a finite group G . Then*

$$|N(H) : H| \equiv |G : H| \pmod{p}$$

PROOF. Let $\mathcal{L}$ be the set of all left cosets of H in G , so that $|\mathcal{L}| = |G : H|$. Let H act on $\mathcal{L}$ by left translation, so that $h * (xH) = (hx)H$. You can easily check that $\mathcal{L}$ becomes an H -set under this action. Let us determine the fixed subset $\mathcal{L}_f$ of $\mathcal{L}$ under the action on H . Notice that $\mathcal{L}_f$ consists on the left cosets of H that are fixed under the action by all the elements of H , namely,

$$\mathcal{L}_f = \{xH \in \mathcal{L} \mid h * (xH) = (hx)H = xH \text{ for all } h \in H\}.$$

Now $(hx)H = xH$ for all $h \in H$ if and only if $x^{-1}hx = x^{-1}h(x^{-1})^{-1} \in H$ for all $h \in H$ if and only if $x^{-1} \in N(H)$ by Lemma 3.1. It shows that the left cosets in $\mathcal{L}_f$ are those contained in $N(H)$. The number of such cosets is $|N(H) : H|$, and so $|N(H) : H| = |\mathcal{L}_f| \equiv |\mathcal{L}| = |G : H| \pmod{p}$ by Lemma 2.18. $\square$

COROLLARY 3.3. *Let H be a p -subgroup of a finite group G . If p divides $|G : H|$, then $N(H) \neq H$.*

PROOF. From Lemma 3.2 we have that p divides $|N(H) : H|$, which yields that $|N(H) : H| \neq 1$, and so $N(H) \neq H$. $\square$

We are now ready to prove the first of the Sylow theorems, which asserts the existence of p -subgroups of a group G for any prime power dividing $|G|$.

THEOREM 3.4. FIRST SYLOW THEOREM

Let G be a finite group with order $|G| = p^n m$, where $n \geq 1$ and the prime p does not divide m . Then

- (i) G contains a subgroup of order p^i for each i where $1 \leq i \leq n$.
- (ii) Every subgroup H of G of order p^i is a normal subgroup of a subgroup of order p^{i+1} for $1 \leq i < n$.

PROOF. (i) We proceed by induction on n . We know that G contains a subgroup of order p by Cauchy's Theorem. Assume that G contains a subgroup H of order p^i , and let us prove the existence of a subgroup of order p^{i+1} . From $i < n$ we derive that p divides $|G : H|$, and by Lemma 3.2 we have that p also divides $|N(H) : H|$. On the other hand, we can consider the factor group $N(H)/H$ because H is normal in $N(H)$. By Cauchy's Theorem, the factor group $N(H)/H$ has a subgroup $\mathcal{K}$ of order p . By the Correspondence Theorem, the set $K = \{g \in N(H) \mid gH \in \mathcal{K}\}$ is a subgroup of $N(H)$, and hence of G , that contains H and such that $\mathcal{K} = K/H$. To finish note that the order of K is p^{i+1} , since $p = |\mathcal{K}| = |K|/p^i$. The result now follows by induction.

(ii) Repeat the construction in (i) and note that $H \leq K \leq N(H)$, where $|K| = p^{i+1}$. Then H is normal in K , since it is normal in $N(H)$. $\square$

Definition 3.5. Let G be a group. A **Sylow p -subgroup** P of G is a maximal p -subgroup of G , that is, a p -subgroup contained in no larger p -subgroup.

If G is a finite group with order $|G| = p^n m$, where $n \geq 1$ and the prime p not dividing m . Then a Sylow p -subgroup of G is nothing more than a subgroup

of G of order p^n . The First Sylow Theorem guarantees the existence of Sylow p -subgroups.

For instance, if G is a group of order $100 = 2^2 \cdot 5^2$, a Sylow 2-subgroup will have order 4, while a Sylow 5-subgroup will have order 25.

Examples 3.6.

1. Consider $\mathbb{Z}_{12}$. Since $12 = 2^2 \cdot 3$, Sylow 2-subgroups will have order 4, and Sylow 3-subgroups will have order 3. More precisely, the only Sylow 2-subgroup of $\mathbb{Z}_{12}$ is $\langle \bar{3} \rangle = \{\bar{0}, \bar{3}, \bar{6}, \bar{9}\}$, and the only Sylow 3-subgroup of $\mathbb{Z}_{12}$ is $\langle \bar{4} \rangle = \{\bar{0}, \bar{4}, \bar{8}\}$.

2. Let $D_3 = \{1, a, a^2, b, ba, ba^2\}$, where $\text{o}(a) = 3$, $\text{o}(b) = 2$, and $aba = b$. The order of D_3 is $6 = 2 \cdot 3$. The only Sylow 3-subgroup of D_3 is $\{1, a, a^2\}$. But there are three Sylow 2-subgroups, namely, $\{1, b\}$, $\{1, ba\}$ and $\{1, ba^2\}$.

If P is a Sylow p -subgroup, then every conjugate gPg^{-1} of P is also a Sylow p -subgroup (the order of P and any of its conjugates is the same). The second Sylow Theorem states that every Sylow p -subgroup arises as this form, that is, any two Sylow p -subgroups are conjugate.

THEOREM 3.7. SECOND SYLOW THEOREM

Let P_1 and P_2 be Sylow p -subgroups of a finite group G . Then P_1 and P_2 are conjugate subgroups of G .

PROOF. Let $\mathcal{L}$ be the collection of left cosets of P_1 , and let P_2 act on $\mathcal{L}$ by $y * (xP_1) = (yx)P_1$ for $y \in P_2$. It is straightforward to check that $\mathcal{L}$ is a P_2 -set. By Lemma 2.18, $|\mathcal{L}_f| \equiv |\mathcal{L}| = |G : P_1| \pmod{p}$. Now p does not divide $|G : P_1|$ because P_1 is a Sylow p -subgroup, and so $|\mathcal{L}_f| \neq 0$. If $xP_1 \in \mathcal{L}_f$, then $yxP_1 = xP_1$ for all $y \in P_2$, so $x^{-1}yx \in P_1$ for all $y \in P_2$. Thus $x^{-1}P_2x \subseteq P_1$. Since $|x^{-1}P_2x| = |P_2| = |P_1|$, we have that $P_1 = x^{-1}P_2x$, which says that P_1 and P_2 are conjugate subgroups. $\square$

COROLLARY 3.8. *A Sylow p -subgroup P of a finite group G is normal in G if and only if P is the unique Sylow p -subgroup of G .*

PROOF. Assume that P is the only Sylow p -subgroup of G , then $gPg^{-1} = P$ because gPg^{-1} is also a Sylow p -subgroup of G , which shows that P is normal. Conversely, assume that $P \triangleleft G$. By the Second Sylow Theorem, any other Sylow p -subgroup P_1 of G must be a conjugate of P , that is, $P_1 = g_1Pg_1^{-1} = P$ because P is normal. Thus P is the only Sylow p -subgroup of G . $\square$

Example 3.9. Let $D_3 = \{1, a, a^2, b, ba, ba^2\}$, where $\text{o}(a) = 3$, $\text{o}(b) = 2$, and $aba = b$. The Sylow 2-subgroups of D_3 are

$$P_1 = \{1, b\}, \quad P_2 = \{1, ba\} \quad \text{and} \quad P_3 = \{1, ba^2\}.$$

By the Second Sylow Theorem we know that P_1 , P_2 and P_3 are conjugates. Let us illustrate this here. The conjugacy class of b is $\text{class } b = \{b, ba, ba^2\}$ and $aba^{-1} = ba$ and $a^2ba^{-2} = ba^2$. Therefore $P_2 = aP_1a^{-1}$ and $P_3 = a^2P_1a^{-2}$.

The last of the Sylow Theorems provides us quite a lot of information about the number of Sylow p -subgroups.

THEOREM 3.10. THIRD SYLOW THEOREM

Let G be a finite group and p a prime dividing $|G|$. Then the number n_p of Sylow p -subgroups of G is congruent to 1 modulo p and divides $|G|$.

PROOF. Let P be a Sylow p -subgroup of G (we know that such a P exists by the First Sylow Theorem). Let $\mathcal{S}$ be the set of all Sylow p -subgroups and let P act on $\mathcal{S}$ by conjugation, so that $x \in P$ carries $T \in \mathcal{S}$ into xTx^{-1} . By Lemma 2.18, $n_p = |\mathcal{S}| \equiv |\mathcal{S}_f| \pmod{p}$. Let us find $\mathcal{S}_f$. If $T \in \mathcal{S}_f$, then $xTx^{-1} = T$ for all $x \in P$. This says that P is a subgroup of $N(T)$, but also T is a subgroup of $N(T)$. On the other hand, P and T are both Sylow p -subgroups of $N(T)$, since they are Sylow p -subgroups of G . Now by the Second Sylow Theorem, P and T are conjugate in $N(T)$. But T is normal in $N(T)$, and so it is its only conjugate in $N(T)$. Thus $P = T$, and we have proved that $\mathcal{S}_f = \{P\}$. From $n_p = |\mathcal{S}| \equiv |\mathcal{S}_f| = 1 \pmod{p}$, we see that the number of Sylow p -subgroups of G is congruent to 1 modulo p .

To finish, let G act on $\mathcal{S}$ by conjugation. The Second Sylow Theorem yields that there is only one orbit in $\mathcal{S}$ under G . If $P \in \mathcal{S}$, then by the Orbit Decomposition Theorem $n_p = |\mathcal{S}| = |\text{class } P| = |G : N(P)|$, where $N(P)$ is

the normalizer of P . Since $|G : N(P)|$ divides $|G|$, we can conclude that the number of Sylow p -subgroups of G divides $|G|$. $\square$

COROLLARY 3.11. *Let G be a finite group of order $p^n m$, where the prime p does not divide m . Then the number n_p of Sylow p -subgroups of G divides m .*

PROOF. From Second Sylow Theorem we know that $n_p \equiv 1 \pmod{p}$ and $n_p || |G|$. Since p does not divide n_p , we have that $n_p | m$. $\square$

3.2 Applications of the Sylow Theorems

In this section we derive some applications of the Sylow Theorems. We will use the Sylow Theorems to show that a group of a particular order can not be prime. More generally, we will learn different techniques to prove that all groups of certain orders are abelian, or that they have nontrivial proper subgroups, that is, they are not simple.

Application 3.12. If p and q are primes, show that no group of order pq is simple.

Solution: Let G be a group of order pq . If $p = q$, then $|G| = p^2$ and we proved in the previous chapter that in such a case G is abelian, and either $G \cong \mathbb{Z}_{p^2}$ or $G \cong \mathbb{Z}_p \times \mathbb{Z}_p$. In any case, this shows that G is not simple.

Assume now that $p \neq q$. Let us say that $p > q$. By the First Sylow Theorem, G has at least one Sylow p -subgroup. On the other hand, if n_p denotes the number of Sylow p -subgroups of G , then by the Third Sylow Theorem we have that $n_p \equiv 1 \pmod{p}$ and $n_p | q$. It implies that $n_p = 1$ because $p > q$, which tells us that there is only one Sylow p -subgroup. This unique Sylow p -subgroup is normal by Corollary 3.8. Thus G is not simple.

Application 3.13. No group of order p^r for $r > 1$ is simple.

Solution: Let G be a group of order p^r , for $r > 1$. By the First Sylow Theorem G contains a subgroup of order p^{r-1} , which is normal in a subgroup of order p^r , that is, the group G . We have then showed that G contains a nontrivial normal subgroup, so G can not be simple. For example, a group of order 16 is not simple because it has a normal subgroup of order 8.

Application 3.14. No group of order 20 is simple.

Solution: Let G be a group of order $20 = 2^2 \cdot 5$. By the First Sylow Theorem G contains Sylow 5-subgroups. Moreover if n_5 denotes the number of Sylow 5-subgroups of G , then by the Third Sylow Theorem $n_5 \equiv 1 \pmod{5}$ and $n_5 | 4$. Thus $n_5 = 1$ and the unique Sylow 5-subgroup is normal in G , and so G is not simple.

Notice that the technique used in all the previous examples has been to show that our group G has only one Sylow p -subgroup for some prime p dividing the order of G . This unique Sylow p -subgroups yields the nonsimplicity of G . Let us use once more time to show that there are no simple groups of order 30.

Application 3.15. No group of order 30 is simple.

Solution: Let G be a group of order 30. Let n_5 and n_3 denote the number of Sylow 5-subgroups and 3-subgroups of G , respectively. The third Sylow Theorem yields that either $n_5 = 1$ or 6, and $n_3 = 1$ or 10. If G has 6 Sylow 5-subgroups, let us say, $P_1, \dots, P_6$. Then $P_i \cap P_j$ (for $i \neq j$) is a subgroup of P_i and of P_j of order dividing 5, and hence just $\{1\}$. Thus each P_i contains 4 elements of order 5 that are not in P_j for $i \neq j$. This gives 24 elements of order 5 in G . If G has 10 Sylow 3-subgroups, then proceeding as before we obtain 10 elements of order 3 in G . Altogether we count 44 elements in G . So necessarily either $n_5 = 1$ or $n_3 = 1$, which shows that G is not simple.

For the last two applications, we need another lemma.

LEMMA 3.16. *Let H and K be finite subgroups of a group G . Then*

$$|HK| = \frac{|H| \cdot |K|}{|H \cap K|}$$

PROOF. Recall that $HK = \{hk \mid h \in H, k \in K\}$. Let $|H| = r$, $|K| = s$, and $|H \cap K| = t$. Notice that HK has at most rs elements because we might have $h_1, h_2 \in H$ and $k_1, k_2 \in K$ such that $h_1k_1 = h_2k_2$. But if that happens, that is, if $h_1k_1 = h_2k_2$, then $x = (h_2)^{-1}h_1 = k_2(k_1)^{-1}$. Now $(h_2)^{-1}h_1 \in H$ and $k_2(k_1)^{-1} \in K$, so $x \in H \cap K$, and $h_2 = h_1x^{-1}$ and $k_2 = xk_1$.

On the other hand, if for $y \in H \cap K$ we let $h_3 = h_1 y^{-1} \in H$ and $k_3 = y k_1 \in K$, then clearly $h_3 k_3 = h_1 k_1$. Thus each element $u \in HK$ can be written in the form hk , for $h \in H$ and $k \in K$, as many times as there are elements of $H \cap K$, i.e., t times. It shows that the number of elements in HK is rs/t , as desired. $\square$

We will use Lemma 3.16 in the following way: a finite group G can not have subgroups H and K of large order having a very small intersection. The reason for that is that the order of HK would exceed the order of G , which clearly makes no sense. For instance, a group of order 24 can not have two subgroups of order 12 and 8 with an intersection of order 2.

Remark 3.17. A subgroup H of index 2 in a finite group G is always normal. The left cosets are H itself, and the coset consisting of all elements in G that they are not in H (remember that the set of left cosets are a partition of G). The right cosets are exactly these two, so left and right cosets coincide and H is normal in G .

Application 3.18. No group of order 48 is simple.

Solution: Let G be a group of order $48 = 2^4 \cdot 3$. By the Third Sylow Theorem G has either one or three Sylow 2-subgroups of order 16. If there is only one Sylow 2-subgroup, proceeding as before, we can conclude that G is not simple.

Suppose that there are three Sylow 2-subgroups, let us say P_1, P_2 and P_3 . If $|P_i \cap P_j| \leq 4$, then by Lemma 3.16 $P_i P_j$ would have at least 64 elements, which is impossible because $|G| = 48$. Thus $P_i \cap P_j$ has order 8. But then $|P_i : P_i \cap P_j| = |P_j : P_i \cap P_j| = 2$, and so $P_i \cap P_j$ is normal in both P_i and P_j by the Sylow First Theorem. Hence the normalizer $N(P_i \cap P_j)$ of $P_i \cap P_j$ contains P_i and P_j , and must have order a multiple > 16 of 16 and a divisor of 48, so 48. Thus $N(P_i \cap P_j) = G$ and $P_i \cap P_j \triangleleft G$.

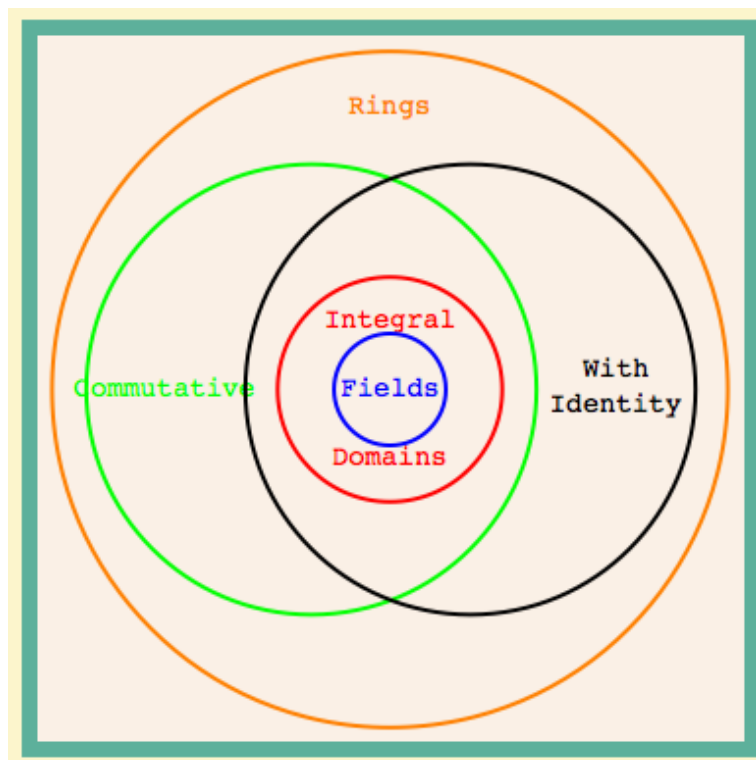
Part II

Rings



Chapter IV

Rings



Chapter 4

Rings

The second part of the course is devoted to the study of rings, which are algebraic systems having two binary operations instead just one. In this first chapter we will follow the same scheme than in the study of groups.

You might be surprise and looking for an example of an algebraic system with two operations. Do they really exist? Yes! ☺ For instance, the sets of numbers $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ have two binary operations: they are closed under addition and multiplication.

4.1 What is a ring?

Definition 4.1. A set R is called a **ring** if it has two binary operations, written as addition and multiplication, satisfying the following axioms for all a , b , and c in R .

R1 $a + b = b + a$.

R2 $a + (b + c) = (a + b) + c$.

R3 There exists an element 0 in R such that $0 + a = a$ for all a .

R4 For each a in R there exists an element $-a$ in R such that $a + (-a) = 0$.

R5 $a(bc) = (ab)c$.

R6 An element 1 in R exists such that $1 \cdot r = r = r \cdot 1$ for all r .

R7 The **distributive laws** hold: $a(b + c) = ab + ac$ and $(b + c)a = ba + ca$.

The ring R is said to be **commutative** if, in addition,

R8 $ab = ba$ for all a and b in R .

Note that the first four axioms assert that $(R, +, 0)$ is an **abelian group**. The additive unity 0 in **R3** is called the **zero** of R , and the additive inverse $-a$ of a in **R4** is called the **negative** of the element a . Thus, we could apply our knowledge on abelian groups to $(R, +, 0)$.

The element 1 , called the **unity** of R , is unique. Be careful! R **might not be a group under the multiplicative operation** because we are not requiring the existence of inverses under the multiplicative operation. Think, for instance, about $\mathbb{Z}$.

In this course we are only going to study rings satisfying axiom **R6**, that is, rings with unity, also called **unital rings**. If we removed **R6** from our list of axioms, R is said to be a **non-unital ring**. The most common example of a non-unital ring is $2\mathbb{Z}$, the set of even integers.

Examples 4.2.

1. Each of $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ is a commutative ring.
2. $\mathbb{Z}_n$ is a commutative ring for each $n \geq 2$.
3. Let $n \geq 1$. The set $M_n(\mathbb{R})$ of all $n \times n$ matrices over $\mathbb{R}$ is a ring using matrix addition and multiplication. Note that $M_n(\mathbb{R})$ is a **noncommutative ring**.

Moreover, if R is a ring, then $M_n(R)$ is a ring under addition and matrix multiplication.

4. The set $\mathbb{R}[x]$ of all polynomials with coefficients in $\mathbb{R}$ is a **commutative ring** with the usual addition and multiplication.
5. If R and S are two rings, then their cartesian product $R \times S$ is a new ring. Recall that $R \times S$ consists of all the ordered pairs (r, s) where r is in R and s is in S . It is easy to check that $R \times S$ is a ring with operations:

$$(r, s) + (r', s') = (r + s, r' + s') \quad (r, s)(r', s') = (rs, r's')$$

The new ring $R \times S$ is called the **direct product** of the rings R and S . Moreover, $R \times S$ is commutative if and only if each R and S are so. The unity is $(1_R, 1_S)$ and the zero is $(0, 0)$.

In general, if $R_1, \dots, R_n$ are rings, then the cartesian product $R_1 \times \dots \times R_n$ is a ring with the following operations:

$$\begin{aligned}(r_1, \dots, r_n) + (s_1, \dots, s_n) &= (r_1 + s_1, \dots, r_n + s_n), \\ (r_1, \dots, r_n)(s_1, \dots, s_n) &= (r_1 s_1, \dots, r_n s_n),\end{aligned}$$

The resulting ring $R_1 \times \dots \times R_n$ is called the **direct product** of the rings $R_1, \dots, R_n$. Notice that $R_1 \times \dots \times R_n$ is commutative if and only if each R_i is commutative. The unity is $(1, \dots, 1)$, and the zero is $(0, \dots, 0)$.

6. The set $R = \{0\}$ is a ring where $0 + 0 = 0$ and $0 \cdot 0 = 0$. It is called the **zero ring** and denoted $R = 0$. Here $1_R = 0$.

We all know that the result of multiplying any number with zero produces zero. It is natural to ask whether this remains true in any ring. Let us take a look to the following result.

THEOREM 4.3. *Let R be a ring, and r and s be arbitrary elements of a ring R . Then*

- (i) $0r = 0 = r0$.
- (ii) $(-r)s = r(-s) = -(rs)$.
- (iii) $(-r)(-s) = rs$.
- (iv) $(mr)(ns) = mn(rs)$ for all integers m and n .

PROOF. (i) $0r = (0 + 0)r = 0r + 0r$, which implies that $0 = 0r$ by adding $-0r$ to both sides (cancellation laws in the group $(R, +, 0)$).

(ii) Apply (i) to get that $(-r)s + rs = (-r + r)s = 0s = 0 = -(rs) + rs$. Hence $(-r)s = -(rs)$ by cancellation. Similarly, $r(-s) = -(rs)$.

(iii) From (ii) we get $(-r)(-s) = r(-(-s)) = rs$, showing (iii).

(iv) Let us first prove that

$$r(ns) = n(rs), \quad \text{for all } n \in \mathbb{Z}. \quad (4.1)$$

This is true for $n = 0$ by (i). Next, if $r(ns) = n(rs)$ for some $n \geq 0$, then

$$r((n+1)s) = r(ns + s) = r(ns) + rs = n(rs) + rs = (n+1)rs.$$

Hence (4.1) holds for every integer $n \geq 0$ by induction. If $n < 0$, write $n = -k$, $k > 0$. Then $r(ns) = r(-(ks)) = -r(ks) = -k(rs) = n(rs)$, which proves (4.1). Similarly, one can prove that

$$(mr)s = m(rs), \quad \text{for all } n \in \mathbb{Z}. \quad (4.2)$$

Given n and m in $\mathbb{Z}$, we have that

$$(mr)(ns) \stackrel{(4.2)}{=} m(r(ns)) \stackrel{(4.1)}{=} m(n(rs)) = (mn)(rs),$$

as desired. $\square$

When dealing with numbers, we learnt that if the product of two numbers is equal to zero, then one of the factors must be zero. We know that this does not work with modular arithmetic. For instance, $\bar{2}$ and $\bar{3}$ are nonzero elements in $\mathbb{Z}_6$, but their product is zero: $\bar{2} \cdot \bar{3} = \overline{2 \cdot 3} = \bar{6} = \bar{0}$. So, it makes sense to introduce the following concept.

Definition 4.4. Let R be a ring. An element a in R is called a **divisor of zero** or a **zero divisor** if a is nonzero and there is a nonzero element b in R such that either $ab = 0$ or $ba = 0$.

In the previous example, $\bar{2}$ and $\bar{3}$ are zero divisors in $\mathbb{Z}_6$. Let us see a more interesting example $\odot$ Consider the ring $\mathbb{M}_2(\mathbb{R})$ of 2×2 matrices with coefficients in $\mathbb{R}$. There are plenty of examples of zero divisors in $\mathbb{M}_2(\mathbb{R})$! For example:

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix},$$

and so $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ are both zero divisors in $\mathbb{M}_2(\mathbb{R})$.

Note that **there are rings which have no zero divisors at all!** The numbers $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$ do not have any zero divisors. Rings without zero divisors are called domains. More precisely,

Definition 4.5. A ring R is a **domain** if $ab = 0$ in R implies that either $a = 0$ or $b = 0$. It means that R does not have any zero divisors. A commutative domain is called an **integral domain**.

The rings $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$ are indeed integral domains. In $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$ we have the **cancellation property**: if either $ab = ac$ or $ba = ca$, then $b = c$. But $\mathbb{Z}_6$ does not satisfy the cancellation property since $\bar{4} \cdot \bar{2} = \bar{4} \cdot \bar{5}$ and $\bar{2} \neq \bar{5}$. Notice that $\bar{4}$ is a zero divisor in $\mathbb{Z}_6$, so it may be a connection between the cancellation property and zero divisors. The next result tells us that our intuition is correct.

THEOREM 4.6. *The following conditions are equivalent for a ring R .*

- (i) R is a domain.
- (ii) If $ab = ac$ in R and $a \neq 0$, then $b = c$.
- (iii) If $ba = ca$ in R and $a \neq 0$, then $b = c$.

PROOF. We will prove that (i) $\Leftrightarrow$ (ii). The proof of (i) $\Leftrightarrow$ (iii) is analogous.

(i) $\Rightarrow$ (ii).

Assume that $ab = ac$ in R , where $a \neq 0$. Then $ab - ac = 0$, so $a(b - c) = 0$. As $a \neq 0$ and R is a domain, we get that $b - c = 0$; that is, $b = c$.

(ii) $\Rightarrow$ (i).

Suppose that $ab = 0$ in R . If $a = 0$ there is nothing to prove. If $a \neq 0$, applying (2) to $ab = a0$ gives $b = 0$. It shows that R is a domain. $\square$

As mentioned before, **not every element in a ring needs to have an inverse under multiplication**. The elements having multiplicative inverses are called **units**. More precisely:

Definitions 4.7. Let R be any ring. An element $u \in R$ is a **unit** if there exists $u' \in R$ such that $uu' = 1 = u'u$. The set of all units in R , denoted R^* or $U(R)$, is a subgroup of $(R, \cdot, 1)$, called the **group of units of the ring R** .

R is a **division ring** if $R \neq \{0\}$ and every nonzero element of R is a unit in R ; that is, $U(R) = R - \{0\}$. A commutative division ring is called a **field**.

Examples 4.8. $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ are examples of fields. From $U(\mathbb{Z}) = \{1, -1\}$, we can conclude that $\mathbb{Z}$ is not a field. It is an integral domain. Other examples of fields are the $\mathbb{Z}_p$ with p prime.

The next results show that the notions of domain, division ring and field are closely related.

THEOREM 4.9. *Every division ring is a domain.*

PROOF. Let R be a division ring. Assume that $ab = 0$ and $a \neq 0$. We need to show that $b = 0$. Since R is a division ring and $a \neq 0$ there exists $a^{-1} \in R$ such that $aa^{-1} = a^{-1}a = 1$. But $0 = a^{-1}(ab) = (a^{-1}a)b = b$, finishing the proof. $\square$

COROLLARY 4.10. *Every field is an integral domain.*

The converse of Corollary 4.10 does not hold, that is, **not every integral domain is a field**. For instance, $\mathbb{Z}$ is an integral domain and $\mathbb{Z}$ is not a field. Nevertheless,

THEOREM 4.11. *Every finite integral domain is a field.*

PROOF. Let R be a finite integral domain. Assume that $|R| = n$ and write $R = \{r_1, \dots, r_n\}$. Given $a \neq 0$ in R , the cardinal of set $aR = \{ar_1, \dots, ar_n\}$ is n (because $ar_i = ar_j$ implies $r_i = r_j$), so since $aR \subseteq R$ and $|R| = n$, we have that $aR = R$. In particular, $1 \in aR$, say $1 = ar_k$ for some $r_k \in R$. Because R is commutative, it shows that a is a unit. Therefore R is a field. $\square$

4.2 Subrings and ideals

We have already seen several examples of rings contained in bigger rings. For example, $\mathbb{Z}$ is a ring inside $\mathbb{Q}$, and $\mathbb{Q}$ is a ring inside $\mathbb{R}$. In such cases, we will say that $\mathbb{Z}$ is a subring of $\mathbb{Q}$, and $\mathbb{Q}$ is a subring of $\mathbb{R}$. Notice that $\mathbb{Z}$ is also a subring of $\mathbb{R}$. Roughly speaking, when a ring S is contained in a larger ring R we say that S is a subring of R . The notion of a subring is the analogue for rings of the notion of subgroups for groups. Here is the formal definition:

Definition 4.12. Let R be a ring, a subset S of R is called a **subring** of R if it is itself a ring with the same operations (including the unity) as R .

Applying the Subgroup Test we get the following procedure to determine whether a subset of a ring is a subring.

THEOREM 4.13. SUBRING TEST.

A subset S of a ring R is a subring if and only if the following conditions are satisfied:

- (i) $0 \in S$ and $1 \in S$.
- (ii) If $s \in S$ and $t \in S$, then $s + t$, st , and $-s$ are all in S .

Let us now see some examples.

Example 4.14. Let R be any ring, and let

$$T_2(R) = \begin{pmatrix} R & R \\ 0 & R \end{pmatrix} = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} : a, b, c \in R \right\}.$$

Then $T_2(R)$ is a subring of $M_2(R)$, called the ring of **upper triangular matrices** over R .

Solution: Clearly, $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ and $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ are in $T_2(R)$. Given $A = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$ and $B = \begin{pmatrix} p & q \\ 0 & r \end{pmatrix}$ in $T_2(R)$, we have that

$$-A = \begin{pmatrix} -a & -b \\ 0 & -c \end{pmatrix}, \quad A + B = \begin{pmatrix} a+p & b+q \\ 0 & c+r \end{pmatrix}, \quad AB = \begin{pmatrix} ap & aq+br \\ 0 & cr \end{pmatrix},$$

are in $T_2(R)$. So, $T_2(R)$ is a subring of $M_2(R)$ by the Subring Test.

Example 4.15. Let R be a ring. Check that the **centre** of R , defined as,

$$Z(R) = \{z \in R \mid zr = rz \text{ for all } r \in R\}$$

is a **subring** of R . Elements in $Z(R)$ are said to be **central** in R .

In our study of groups, we spent some time studying certain special subgroups called normal subgroups. If subrings play the same role in rings than subgroups in groups, what will the counterpart of normal subgroups be?

Definition 4.16. A subset I of a ring R is called an **ideal** of R if I is a **subgroup** of $(R, +, 0)$ and $Ra \subseteq I$, $aR \subseteq I$, for every $a \in I$.

Ideals are in rings as normal subgroups are in groups.

Example 4.17. The set $I = \begin{pmatrix} 0 & \mathbb{R} \\ 0 & \mathbb{R} \end{pmatrix}$ is an ideal of $T_2(\mathbb{R})$.

- I is a subgroup of $(T_2(\mathbb{R}), +)$: $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \in I$ and

$$\begin{pmatrix} 0 & a_{12} \\ 0 & a_{22} \end{pmatrix} + \begin{pmatrix} 0 & b_{12} \\ 0 & b_{22} \end{pmatrix} = \begin{pmatrix} 0 & a_{12} + b_{12} \\ 0 & a_{22} + b_{22} \end{pmatrix}, \quad -\begin{pmatrix} 0 & a_{12} \\ 0 & a_{22} \end{pmatrix} = \begin{pmatrix} 0 & -a_{12} \\ 0 & -a_{22} \end{pmatrix}$$

are elements in I .

- $IT_2(\mathbb{R}) \subseteq I$: $\begin{pmatrix} 0 & a_{12} \\ 0 & a_{22} \end{pmatrix} \begin{pmatrix} c_{11} & c_{12} \\ 0 & c_{22} \end{pmatrix} = \begin{pmatrix} 0 & a_{12}c_{22} \\ 0 & a_{22}c_{22} \end{pmatrix}$
- $T_2(\mathbb{R})I \subseteq I$: $\begin{pmatrix} c_{11} & c_{12} \\ 0 & c_{22} \end{pmatrix} \begin{pmatrix} 0 & a_{12} \\ 0 & a_{22} \end{pmatrix} = \begin{pmatrix} 0 & c_{11}a_{12} + c_{12}a_{22} \\ 0 & c_{22}a_{22} \end{pmatrix}$

Principal and finitely generated ideals

Let R be a ring and A any subset of R . The **ideal generated by** A , denoted $\langle A \rangle$, is the smallest ideal of R containing A . An ideal I of R is

- **principal** if I is generated for a single element, that is, $I = \langle a \rangle$. In this case, $A = \{a\}$ but we drop the set brackets in the notation.
- **finitely generated** if I is generated for a finite set, that is, $I = \langle A \rangle$ for $A = \{a_1, \dots, a_n\}$. In this case, again we drop the set brackets and we simply write $I = \langle a_1, \dots, a_n \rangle$.

Keeping in mind that the intersection of ideals is again an ideal of R , one can easily check that

$$\langle A \rangle = \cap_{I \triangleleft R, A \subseteq I} I$$

That is, $\langle A \rangle$ is the intersection of all the ideals of R that contains the set A . When R is commutative

$$\langle A \rangle = RA = AR = RAR,$$

where RA denote the set of all finite sums of elements of the form ra with $r \in R$ and $a \in A$, that is, $RA = \{r_1a_1 + r_2a_2 + \dots + r_na_n \mid r_i \in R, a_i \in A, n \in \mathbb{Z}^+\}$. Similarly, $AR = \{a_1r_1 + a_2r_2 + \dots + a_nr_n \mid r_i \in R, a_i \in A, n \in \mathbb{Z}^+\}$, and

$$RAR = \{r_1 a_1 s_1 + r_2 a_2 s_2 + \dots + r_n a_n s_n \mid r_i, s_i \in R, a_i \in A, n \in \mathbb{Z}^+\}.$$

For the particular case of principal ideals, when either R is **commutative** and $a \in R$, or R is **not necessarily commutative but** $a \in Z(R)$, the principal ideal $\langle a \rangle$ generated by a is nothing more than the set of all R -multiples of a , i.e.,

$$\langle a \rangle = Ra = \{ra \mid r \in R\} = \{ar \mid r \in R\} = aR$$

If R is not commutative and a is not necessarily a central element, then

$$\langle a \rangle = \{r_1 a s_1 + r_2 a s_2 + \dots + r_n a s_n \mid r_i, s_i \in R, n \in \mathbb{Z}^+\}$$

Constructing principal ideals in a commutative ring is a similar procedure to the one of generating cyclic subgroups of a group. Notice that an element $b \in \langle a \rangle$ if and only if $b = ra$ for some $r \in R$ if and only if b is a multiple of a or, “ a divides b in R ”. Also $b \in \langle a \rangle$ if and only if $\langle b \rangle \subseteq \langle a \rangle$.

Examples 4.18.

1. The trivial ideals 0 and R are both principal: $0 = \langle 0 \rangle$ and $R = \langle 1 \rangle$.
2. In $\mathbb{Z}$ we have $n\mathbb{Z} = \mathbb{Z}n = \langle n \rangle = \langle -n \rangle$ for all integers n . These are the only ideals of $\mathbb{Z}$ since in particular they are the only subgroups of $\mathbb{Z}$. So every ideal of $\mathbb{Z}$ is principal. The ideal generated by two nonzero integers n and m is the principal ideal generated by their greatest common divisor d , that is, $\langle n, m \rangle = \langle d \rangle$.

3. The ideal $\langle 2, x \rangle$ generated by 2 and x in $\mathbb{Z}[x]$ is not a principal ideal. By definition $\langle 2, x \rangle = \{2p(x) + xq(x) \mid p(x), q(x) \in \mathbb{Z}[x]\}$, and so $\langle 2, x \rangle$ consists of the polynomials in $\mathbb{Z}[x]$ whose constant term is even.

Assume on the contrary that $\langle 2, x \rangle = \langle a(x) \rangle$ for some $a(x) \in \mathbb{Z}[x]$. Since $2 \in \langle 2, x \rangle$, we must have that $2 = p(x)a(x)$ for $p(x) \in \mathbb{Z}[x]$. Now, $0 = \deg(2) = \deg(pa) = \deg(p) + \deg(a)$, which yields that $p(x)$ and $a(x)$ must both be constant polynomials, i.e., integers. But 2 is prime, so $a(x), p(x) \in \{\pm 1, \pm 2\}$. If $a(x) = \pm 1$, then $\langle a(x) \rangle = \mathbb{Z}[x]$ but this is a contradiction because obviously $\langle 2, x \rangle \neq \mathbb{Z}[x]$. The only possibility is $a(x) = \pm 2$, but $x \in \langle a(x) \rangle = \langle 2 \rangle$ and so $x = 2q(x)$ for some $q(x) \in \mathbb{Z}[x]$, which is impossible. Thus $\langle 2, x \rangle$ is not principal.

The next result is quite interesting. It says that the only possible ideal containing a unit is the ring itself.

THEOREM 4.19. *The following are equivalent for an ideal I of a ring R .*

- (i) $1 \in I$.
- (ii) I contains a unit.
- (iii) $I = R$.

PROOF. (i) $\Rightarrow$ (ii) and (iii) $\Rightarrow$ (i) are obvious.

(ii) $\Rightarrow$ (iii). Let $u \in R$ be a unit. Then $1 = u^{-1}u \in I$ because I is an ideal of R . Hence, $r = r \cdot 1 \in I$ for all $r \in R$, proving (iii). $\square$

4.3 Homomorphism and Isomorphism

Recall that the group homomorphisms, the structure-preserving mappings from one group to another, turned out to be just those that preserved the group operation. Ring homomorphisms are almost the same for rings as for groups. Since rings have two operations, a map $\theta : R \rightarrow S$, where R and S are rings, should be structure-preserving if θ preserves both addition and multiplication. However, in a ring R , the unity 1_R is also part of the structure, so we require θ to preserve the unity: $\theta(1_R) = 1_S$.

Definition 4.20. Let R and S be rings, a mapping $\theta : R \rightarrow S$ is called a **ring homomorphism** if, for all r and r' in R :

(1) θ **preserves addition:** $\theta(r + r') = \theta(r) + \theta(r')$

(2) θ **preserves multiplication:** $\theta(rr') = \theta(r)\theta(r')$

(3) θ **preserves the unity:** $\theta(1_R) = 1_S$

θ is a **monomorphism** (respectively, **epimorphism**) if θ is injective (respectively, surjective). Bijective homomorphisms are called **isomorphisms**. If there is an epimorphism from R to S , we call S a **homomorphic image of R** . If there is an isomorphism θ from R to S , we say that R is **isomorphic to S** and we write $R \cong S$, or $R \overset{\theta}{\cong} S$.

Example 4.21. The projections from the direct product of rings onto each of the components are examples of epimorphisms. More precisely, if R and S are rings, then the projections $\pi_1 : R \times S \rightarrow R$ and $\pi_2 : R \times S \rightarrow S$ are ring epimorphisms, where $\pi_1(r, s) = r$ and $\pi_2(r, s) = s$.

Example 4.22. There are several ways of **constructing the complex numbers**. Here, we will see them as a subring of the ring of square matrices with coefficients in $\mathbb{R}$. Let us start by noticing that

$$R = \left\{ \begin{pmatrix} a & -b \\ b & a \end{pmatrix} : a, b \in \mathbb{R} \right\}$$

is a ring (a subring of $M_2(\mathbb{R})$) under matrix addition and matrix multiplication. Our goal is to show that R and $\mathbb{C}$ are isomorphic. To do so, we need to find a bijective map between R and $\mathbb{C}$ satisfying relations (1) - (3) above.

We try with the map $\theta : R \rightarrow \mathbb{C}$ given by $\theta \begin{pmatrix} a & -b \\ b & a \end{pmatrix} = a + bi$. Clearly, θ is bijective (complete the details yourself). Take $r = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$, $r' = \begin{pmatrix} a' & -b' \\ b' & a' \end{pmatrix}$ in R .

(1) θ preserves addition:

$$\begin{aligned} \theta(r + r') &= \theta \begin{pmatrix} a + a' & -b - b' \\ b + b' & a + a' \end{pmatrix} = (a + a') + (b + b')i = (a + bi) + (a' + b'i) = \\ &= \theta(r) + \theta(r') \end{aligned}$$

(2) θ preserves multiplication:

$$\begin{aligned} \theta(rr') &= \theta \begin{pmatrix} aa' - bb' & -ab' - ba' \\ ba' + ab' & aa' - bb' \end{pmatrix} = (aa' - bb') + (ab' + ba')i = (a + bi)(a' + b'i) = \\ &= \theta(r)\theta(r') \end{aligned}$$

(3) θ preserves the unity: $\theta \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1 + 0i = 1_{\mathbb{C}}$.

Therefore, we can conclude that $R = \left\{ \begin{pmatrix} a & -b \\ b & a \end{pmatrix} : a, b \in \mathbb{R} \right\} \stackrel{\theta}{\cong} \mathbb{C}$

In our groups journey we learnt that there is a subring and an ideal associated to any group homomorphism, namely the range and the kernel. Guess what? We have a similar result for rings! ☺

THEOREM 4.23. *Let $\theta : R \rightarrow S$ be a ring homomorphism.*

- (i) $\theta(R)$ is a subring of S .
- (ii) $\ker \theta = \{r \in R \mid \theta(r) = 0_S\}$ is an ideal of R .

Moreover, θ is a monomorphism if and only if $\ker \theta = 0$.

PROOF. (i) Since θ is an additive group homomorphism, we know that $\theta(R)$ is an additive subgroup of S . Next $\theta(R)$ is closed under multiplication because $\theta(r)\theta(r') = \theta(rr') \in \theta(R)$, and $1_S \in \theta(R)$, since $1_S = \theta(1_R) \in \theta(R)$.

(ii) Applying groups knowledge, we have that $\ker \theta$ is an additive subgroup of R . If $r \in R$ and $a \in \ker \theta$, then $\theta(ra) = \theta(r)\theta(a) = \theta(r)0 = 0$. Thus $ra \in \ker \theta$. Similarly, $ar \in \ker \theta$. Hence $\ker \theta$ is an ideal of R . $\square$

4.4 Factor rings

In our study of groups, we found out a way of manufacturing all the homomorphic images of any group G . We constructed the factor groups of G and we showed that every factor group of G is a homomorphic image of G . Our goal is to reach a similar result for rings. To do so, in this section, we construct factor rings. We begin by defining cosets of rings:

Let R be a ring and I an ideal of R . For any element a in R , the symbol $I + a$ denotes the set of all sums $y + a$, as a remains fixed and y ranges over I . That is,

$$I + a = \{y + a : y \in I\}$$

We call $I + a$ a **coset** of I in R .

Let us ignore for a moment the multiplicative operation of R and focus only on the additive one. We know that R is an abelian group under the addition and I is a subgroup of R . Thus the cosets we have just defined are nothing more than the cosets of the subgroup I of R , with additive notation. Therefore, all the results proved for groups remain valid in the present case. (Be careful! Here we are using additive notation.) For instance, we know

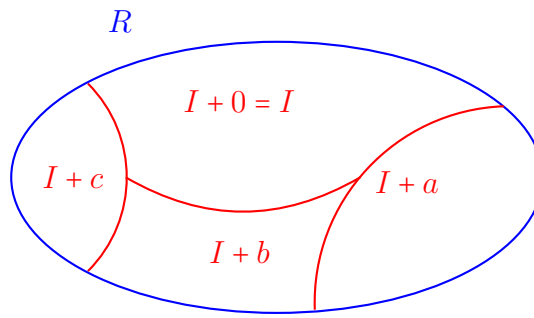
that

$$a \in I + b \Leftrightarrow I + a = I + b \quad (\text{RC1})$$

$$I + a = I + b \Leftrightarrow a - b \in I \quad (\text{RC2})$$

$$I + a = I \Leftrightarrow a \in I \quad (\text{RC3})$$

In particular, from (RC3) we get that $I = I + 0$. Moreover, the family of all cosets of $I + a$, as a ranges over R , is a partition of R .



As expected, we can add and multiply cosets, following the rules:

$$\begin{aligned} (I + a) + (I + b) &= I + (a + b) \\ (I + a)(I + b) &= I + ab \end{aligned}$$

Put it this way, the sum of the coset of a and the coset of b is the coset of $a + b$; the product of the coset of a and the coset of b is the coset of ab . As in groups, we need to make sure that these coset operations are well defined, in the sense that they should be determined without any ambiguity. More precisely, if $I + a$ is the same coset as $I + c$, and $I + b$ is the same coset of $I + d$, we should check that $I + (a + b)$ is the same coset as $I + (c + d)$ and $I + ab$ is the same coset as $I + cd$. Graphically:

$$\begin{array}{ccc} (I + a) + (I + b) = I + (a + b) & & (I + a)(I + b) = I + ab \\ \parallel & \parallel & \parallel \\ (I + c) + (I + d) = I + (c + d) & \text{and} & (I + c)(I + d) = I + cd \end{array}$$

As we will see in the next result, things work pretty well because we have chosen I to be an ideal of R .

THEOREM 4.24. *Let I be an ideal of a ring R . If $I + a = I + c$ and $I + b = I + d$, then $I + (a + b) = I + (c + d)$ and $I + ab = I + cd$.*

PROOF. We first show that $I + (a + b) = I + (c + d)$. From (RC2) we know that $a - c$ and $b - d$ are in I . Since I is closed under addition, we have that $(a - c) + (b - d) = (a + b) - (c + d)$ is in I . Then $I + (a + b) = I + (c + d)$ by (RC2).

On the other hand, since RI and IR are contained in I , we have that

$$(a - c)b \in I \quad \text{and} \quad c(b - d) \in I$$

Thus $(a - c)b + c(b - d) = ab - cb + cb - cd = ab - cd$ is in I , and $I + ab = I + cd$ by (RC2). $\square$

Now, we consider the set R/I consisting of all cosets of I in R :

$$R/I = \{I + a, I + b, I + c, \dots\} = \{I + a : a \in R\}$$

Observe that we have just endowed R/I with two binary operations. Our next aim is to show that R/I is indeed a ring with the coset addition and coset multiplication.

THEOREM 4.25. *Let I be an ideal of a ring R . Then R/I is a ring with the coset addition and coset multiplication.*

PROOF. It is straightforward to check that coset addition and multiplication are associative, and multiplication is distributive over addition. The zero element of R/I is the coset $I = I + 0_R$, and the negative of $I + a$ is $I + (-a)$. The unity of R/I is $I + 1_R$. $\square$

The ring R/I is called the **factor ring** of R by I . It is easy to check that the factor ring R/I is commutative provided R is so. We are now ready to connect factor rings with homomorphism:

THEOREM 4.26. *Let I be an ideal of a ring R . Then the factor ring R/I is a homomorphic image of R .*

PROOF. Inspired by our result for groups, we consider the map $\pi : R \rightarrow R/I$ which carries every element of R to its own coset, that is, $\pi(a) = I + a$. It is obvious that π is surjective and it can be easily verified that π is a ring homomorphism. $\square$

As for groups, every ideal I of a ring R is the kernel of some ring homomorphism with domain R . One just needs to consider the coset map $\pi : R \rightarrow R/I$ defined by $\pi(a) = I + a$.

Example 4.27. Let $\mathbb{Z}$ be the ring of the integers, and let $\langle 6 \rangle = 6\mathbb{Z}$ be the ideal of $\mathbb{Z}$ consisting of all the multiples of 6. The elements of this group are the cosets of the ideal $6\mathbb{Z}$ in $\mathbb{Z}$, namely:

$$6\mathbb{Z} + 0 = \{\dots, -18, -12, -6, 0, 6, 12, 18, \dots\} = \bar{0}$$

$$6\mathbb{Z} + 1 = \{\dots, -17, -11, -5, 1, 7, 13, 19, \dots\} = \bar{1}$$

$$6\mathbb{Z} + 2 = \{\dots, -16, -10, -4, 2, 8, 14, 20, \dots\} = \bar{2}$$

$$6\mathbb{Z} + 3 = \{\dots, -15, -9, -3, 3, 9, 15, 21, \dots\} = \bar{3}$$

$$6\mathbb{Z} + 4 = \{\dots, -14, -8, -2, 4, 10, 16, 22, \dots\} = \bar{4}$$

$$6\mathbb{Z} + 5 = \{\dots, -13, -7, -1, 5, 11, 17, 23, \dots\} = \bar{5}$$

The rules for adding and multiplying cosets give us the following tables:

+	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$		$\cdot$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$
$\bar{0}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$		$\bar{0}$	$\bar{0}$	$\bar{0}$	$\bar{0}$	$\bar{0}$	$\bar{0}$	$\bar{0}$
$\bar{1}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$	$\bar{0}$		$\bar{1}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$
$\bar{2}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$	$\bar{0}$	$\bar{1}$		$\bar{2}$	$\bar{0}$	$\bar{2}$	$\bar{4}$	$\bar{0}$	$\bar{2}$	$\bar{4}$
$\bar{3}$	$\bar{3}$	$\bar{4}$	$\bar{5}$	$\bar{0}$	$\bar{1}$	$\bar{2}$		$\bar{3}$	$\bar{0}$	$\bar{3}$	$\bar{0}$	$\bar{3}$	$\bar{0}$	$\bar{3}$
$\bar{4}$	$\bar{4}$	$\bar{5}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$		$\bar{4}$	$\bar{0}$	$\bar{4}$	$\bar{2}$	$\bar{0}$	$\bar{4}$	$\bar{2}$
$\bar{5}$	$\bar{5}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$		$\bar{5}$	$\bar{0}$	$\bar{5}$	$\bar{4}$	$\bar{3}$	$\bar{2}$	$\bar{1}$

Does it look familiar to you? Again, the factor ring $\mathbb{Z}/6\mathbb{Z}$ is isomorphic to the ring $\mathbb{Z}_6$! In general, the ring $\mathbb{Z}_n$ is isomorphic to the factor ring $\mathbb{Z}/n\mathbb{Z}$ and we can consider $\mathbb{Z}_n$ to be the same as $\mathbb{Z}/n\mathbb{Z}$.

We finish this section by describing all the ideals of a factor ring R/I . The following result is the analogue of the Correspondence Theorem for groups.

THEOREM 4.28. *Let I be an ideal of a ring R .*

- (i) *If J is an ideal of R such that $I \subseteq J$, then the set $J/I = \{I + a : a \in J\}$ is an ideal of the factor ring R/I .*

(ii) If $\tilde{J}$ is an ideal of R/I then $\tilde{J} = J/I$ for some (unique) ideal J of R with $I \subseteq J$. In fact, $J = \{x \in R \mid I + x \in \tilde{J}\}$.

(iii) If J and J' are ideals of R , that contain I , then

$$J \subseteq J' \quad \text{if and only if} \quad J/I \subseteq J'/I.$$

PROOF. (i) From the Correspondence Theorem for groups we know that J/I is a subgroup of $(R/I, +, 0_{R/I})$. It remains to check that $(R/I)(I + a)$ and $(R/I)(I + a) \subseteq J/I$, for all $a \in J$. Given $a \in J$ and $r \in R$ we have that $ra, ar \in J$ (because $J \triangleleft R$), which implies that

$$(I + r)(I + a) = I + ra \in J/I, \quad (I + a)(I + r) = I + ar \in J/I.$$

Thus $J/I \triangleleft R/I$.

(ii) Let $\tilde{J}$ be an ideal of R/I , and define $J = \{x \in R \mid I + x \in \tilde{J}\}$. Check that J is an ideal of R . Notice that $I \subseteq J$ because $I + y = I + 0 \in \tilde{J}$ for all $y \in I$. It remains to show that $\tilde{J} = J/I$. We have that $J/I \subseteq \tilde{J}$ because if $I + r \in J/I$ we have that $I + r = I + x$ for some $x \in J$. But $I + x \in \tilde{J}$ because $x \in J$, that is, $I + r \in \tilde{J}$. Conversely, $I + x \in \tilde{J}$ implies that $x \in J$, by the definition of J , and so $I + x \in J/I$.

(iii) follows from the Correspondence Theorem for groups. $\square$

4.5 The Isomorphism Theorem

In the previous section, we showed that any factor ring R/I is a homomorphic image of R . Thus the factor ring construction provides us a way of producing homomorphic images of any ring R . In what follows, we will prove that this is a way of producing all the homomorphic images of R . In other words, any homomorphic image of R can be seen as a factor ring of R . This result is known as the **FIRST ISOMORPHISM THEOREM**.

THEOREM 4.29. FIRST ISOMORPHISM THEOREM. Let $\theta : R \rightarrow S$ be a ring epimorphism and let K be the kernel of θ . Then $S \cong R/K$.

PROOF. To show that the factor ring R/K is isomorphic to the ring S , we need to find an isomorphism from R/K to S . Proceeding in the same way that for groups, we define $\tilde{\theta} : R/K \rightarrow S$ by $\tilde{\theta}(K + x) = \theta(x)$. Ignoring the multiplicative operation and considering R and S as additive groups, θ is a group homomorphism. Thus we can apply the Isomorphism Theorem for groups to conclude that the additive groups R/K and S are isomorphic via $\tilde{\theta}$. In other words, $\tilde{\theta}$ is a group isomorphism from R/K to S . It remains to show that $\tilde{\theta}$ preserves the multiplication and the unity. If $K + a$ and $K + b$ are two elements in R/K , then

$$\tilde{\theta}((K + a)(K + b)) = \tilde{\theta}(K + ab) = \theta(ab) = \theta(a)\theta(b) = \tilde{\theta}(K + a)\tilde{\theta}(K + b)$$

Applying that θ preserves the unity we obtain that $\tilde{\theta}(K + 1_R) = \theta(1_R) = 1_S$, finishing the proof. $\square$

As it happened for groups, the Isomorphism Theorem for rings is very useful. In what follows, we present some applications. For instance, imagine that we are asked to prove that a subset I of a ring R is an ideal of R , and that the factor ring R/I is isomorphic to another ring S . The way to do this is to find a ring epimorphism from R to S with kernel equal to I . Then I is an ideal of R because it is a kernel, and $R/I \cong S$ by the First Isomorphism Theorem. Let us see an example:

Example 4.30. Let I be an ideal of a ring R , show that $M_n(I)$ is an ideal of $M_n(R)$ such that $M_n(R)/M_n(I) \cong M_n(R/I)$.

Solution: Following the technique explained above, all we need to do is to find a ring epimorphism from $M_n(R)$ to $M_n(R/I)$ with kernel $M_n(I)$. Then the result will follow by the First Isomorphism Theorem.

Recall that the coset map $\pi : R \rightarrow R/I$ given by $\pi(a) = I + a$, is a ring epimorphism. Keeping this in mind, we define $\theta : M_n(R) \rightarrow M_n(R/I)$ as $\theta(a_{ij}) = (\pi(a_{ij})) = (I + a_{ij})$. It is straightforward to check that θ is a ring epimorphism. Moreover, $\ker \theta = M_n(I)$ because $\ker \pi = I$. Thus $M_n(I)$ is an ideal of $M_n(R)$ and $M_n(R)/M_n(I) \cong M_n(R/I)$ by the First Isomorphism Theorem.

Like in groups we have a Second and a Third Isomorphism Theorems. We will prove the Second Isomorphism Theorem in the tutorial session.

THEOREM 4.31. SECOND ISOMORPHISM THEOREM. *Let R be a ring, I an ideal of R and S a subring of R . Then $S + I = \{s + y \mid s \in S, y \in I\}$ is a subring of R , I and $I \cap S$ are ideals of $S + I$ and S , respectively, and $(S + I)/I \cong S/(S \cap I)$.*

THEOREM 4.32. THIRD ISOMORPHISM THEOREM.

If $I \subseteq J \subseteq R$, where I and J are ideals of a ring R . Then

$$J/I = \{I + x \mid x \in J\} \triangleleft R/I,$$

and $(R/I)/(J/I) \cong R/J$

PROOF. Let $I \subseteq J \subseteq R$, for $I, J \triangleleft R$. From Theorem 4.28 we have that $J/I \triangleleft R/I$. So, it makes sense to consider the factor ring $(R/I)/(J/I)$. The proof of the Third Isomorphism Theorem for groups tells us that the map $\theta : R/I \rightarrow R/J$ defined by $\theta(I + r) = J + r$ is a group epimorphism with kernel J/I . So we only need to check here that θ preserves multiplication. If $I + r, I + r' \in R/I$, we have that

$$\theta((I + r)(I + r')) = \theta(I + rr') = J + rr' = (J + r)(J + r') = \theta(J + r)\theta(J + r'),$$

which shows that θ is a ring homomorphism. The result now follows from an application of the First Isomorphism Theorem for rings. $\square$

4.6 Simple rings. Prime and maximal ideals

By analogy with groups, a ring R is called a **simple ring** if $R \neq \{0\}$ and the only ideals of R are $\{0\}$ and R . Examples of simple rings are the division rings. In fact, let R be a division ring. If I is a nonzero ideal of R , then we can find a nonzero element y in I . But then y is a unit because R is a division ring. Thus, $I = R$ by an application of Theorem 4.19.

Observe that fields are simple rings since they are division rings. The next result tells us that the converse is true provided that our simple ring is commutative.

THEOREM 4.33. *Let $R \neq 0$ be a commutative ring. Then R is simple if and only if R is a field.*

PROOF. We have already showed that every field is a simple ring. Let R be a commutative simple ring. If $0 \neq a \in R$, then Ra is a nonzero ideal of R and so $Ra = R$. Thus $1 \in Ra$ and $1 = ba = ab$ for some $b \in R$. Hence, a is a unit in R , so R is a field. $\square$

Definitions 4.34. Let R be a ring and I an ideal of R . We say that I is **proper** if $I \neq R$.

A proper ideal I is said to be **prime** if for any two elements a and b in R ; if $ab \in I$, then $a \in I$ or $b \in I$.

A proper ideal J of R is called **maximal** if whenever we have $J \subseteq K \subseteq R$, for $K \triangleleft R$, then either $K = J$ or $K = R$.

In what follows, we will characterize prime and maximal ideals.

THEOREM 4.35. *Let R be a commutative ring, an ideal $I \neq R$ of R is prime if and only if the factor ring R/I is an integral domain.*

PROOF. Observe that R/I is a commutative ring because R is so. If R/I is an integral domain, and $ab \in I$ then $(I + a)(I + b) = I + ab = I$, which is the zero of the factor ring R/I . So $I + a = I$ or $I + b = I$. Hence $a \in I$ or $b \in I$, which shows that I is a prime ideal. Conversely, if I is a prime ideal, let $(I + a)(I + b) = I$, the zero of R/I ; that is, $I + ab = I$. Hence, $ab \in I$, so $a \in I$ or $b \in I$ because I is prime. Thus $I + a = I$ or $I + b = I$, proving that R/I is an integral domain. $\square$

THEOREM 4.36. *Let J be a proper ideal of a ring R . Then J is maximal if and only if R/J is a simple ring.*

PROOF. Assume first that J is a maximal ideal of R . Then $J \neq R$ and $R/J \neq \{0\}$. If K/J is a nonzero ideal of R/J , then K is an ideal of R such that $K \neq J$ and $K \subseteq R$. Thus $K = R$ since J is maximal, and so $K/J = R/J$. Therefore, R/J is simple.

Conversely, suppose that R/J is simple. If K is an ideal of R such that $J \subseteq K \subseteq R$, then K/J is an ideal of R/J . Thus $K/J = 0$ or $K/J = R/J$,

which implies that $K = J$ or $K = R$. It shows that J is a maximal ideal of R , finishing the proof. $\square$

COROLLARY 4.37. *Let R be a commutative ring. Then an ideal I of R is maximal if and only if R/I is a field.*

COROLLARY 4.38. *Every maximal ideal of a commutative ring is a prime ideal.*

The converse of Corollary 4.38 is false! In the ring $\mathbb{Z}$ of integers, the zero ideal is prime by Theorem 4.35, but it is not maximal. If it were, then $\mathbb{Z}$ would be a field by Corollary 4.37 which is not true.

4.7 Fields of Fractions

It is easy to check that every subring of a field is an integral domain. The most surprising fact is that the converse also holds:

Every integral domain is isomorphic to a subring of a field.

For instance, we can view $\mathbb{Z}$ inside $\mathbb{Q}$ by identifying the integer n with the fraction $\frac{n}{1}$. We could do something more general: if D is an integral domain, we can construct a field, denoted $Q(D)$ or Q , called the **field of fractions** of D , such that D can be identified with a subring of Q . We call Q the field of fractions of D because its elements are all “fractions or quotients” $\frac{a}{b}$ where a and b are both in R , $b \neq 0$.

To motivate the definition of Q , let us analyze what happens in the case of $\mathbb{Z}$ and $\mathbb{Q}$. Recall that all the fractions

$$\frac{2}{3}, \frac{4}{6}, \frac{-6}{-9}, \dots$$

are equal. The fraction $\frac{2}{3}$ is the representative of the set consisting of all the pair of integers (m, n) satisfying that $\frac{m}{n} = \frac{2}{3}$. In a more formal way, we could see the fraction $\frac{2}{3}$ as the representative of the class in certain equivalence relation in the set $\{(m, n) \mid m, n \in \mathbb{Z}, n \neq 0\}$. To find out such an equivalence relation just remember that

$$\frac{m}{n} = \frac{m'}{n'} \text{ in } \mathbb{Q} \text{ if and only if } mn' = m'n$$

At this point, we are in a position to construct the field of fractions of an integral domain $\ominus$. Let D be an integral domain and consider the set

$$X = \{(a, b) : a, b \in D, b \neq 0\}$$

We define a binary relation on X by

$$(a, b) \sim (c, d), \quad \text{if and only if} \quad ad = cb.$$

LEMMA 4.39. $\sim$ is an equivalence relation on X .

PROOF. It is clear that $\sim$ is reflexive and symmetric. To show that it is transitive, let $(a, b) \sim (c, d)$ and $(c, d) \sim (e, f)$. Then $ad = cb$ and $cf = ed$, so $(af)d = (ad)f = (cb)f = (cf)b = (ed)b = (eb)d$ because R is commutative. From $(af - eb)d = 0$, $d \neq 0$, we derive $af = eb$, which says $(a, b) \sim (e, f)$. $\square$

Motivated by the case $D = \mathbb{Z}$, we define the **quotient or fraction** $\frac{a}{b}$ to be the equivalence class $[a, b]$ of the pair (a, b) in X . More precisely:

$$\frac{a}{b} = [a, b]$$

Using this notation we have that

$$\frac{a}{b} = \frac{c}{d} \Leftrightarrow [a, b] = [c, d] \Leftrightarrow (a, b) \sim (c, d) \Leftrightarrow ad = cb$$

In particular:

$$\frac{a}{b} = \frac{ra}{rb} \quad \text{for all } r \neq 0 \text{ in } D$$

Notice that we have already constructed the set of fractions that we wanted:

$$Q = \left\{ \frac{a}{b} : a, b \in D, b \neq 0 \right\}$$

Inspired by the addition and multiplication in $\mathbb{Q}$, we define addition and multiplication in Q by

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + cb}{bd} \quad \text{and} \quad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

Notice that $ad + cb$, ac and bd are the operations in D . On the other hand $bd \neq 0$ because b and d are two nonzero elements in an integral domain. We

need to check that the operations above are well-defined in the sense that they do not depend on the representative of the equivalence class. We check it here for addition. As an exercise, you can do the same for the multiplication. If $\frac{a}{b} = \frac{a'}{b'}$ and $\frac{c}{d} = \frac{c'}{d'}$, then we must show that

$$\frac{ad + cb}{bd} = \frac{a'd' + c'b'}{b'd'}, \text{ that is, } (ad + cb)(b'd') = (a'd' + c'b')(bd)$$

From $\frac{a}{b} = \frac{a'}{b'}$ and $\frac{c}{d} = \frac{c'}{d'}$ we have that $ab' = a'b$ and $cd' = c'd$. So

$$\begin{aligned} (ad + cb)(b'd') &= (ab')(dd') + (cd')(db') = (a'b)(dd') + (c'd)(bb') = \\ &= (a'd')(bd) + (c'b')(bd) = (a'd' + c'b')(bd), \end{aligned}$$

as required. It is easy to check that with these two operations Q becomes a field. Notice that the **zero** is $\frac{0}{1}$, the **negative** of $\frac{a}{b}$ is $\frac{-a}{b}$, while the **unity** is $\frac{1}{1}$ and the inverse of $\frac{a}{b}$ is $\frac{b}{a}$, because

$$\frac{a}{b} + \frac{0}{1} = \frac{a1 + 0b}{b1} = \frac{a}{b}$$

$$\frac{a}{b} + \frac{-a}{b} = \frac{ab + (-a)b}{b^2} = \frac{0}{b^2} = \frac{0}{1} \quad \text{because} \quad [0, b^2] = [0, 1]$$

$$\frac{a}{b} \cdot \frac{1}{1} = \frac{a1}{b1} = \frac{a}{b}, \quad \frac{a}{b} \cdot \frac{b}{a} = \frac{ab}{ab} = \frac{1}{1} \quad \text{because} \quad [ab, ab] = [1, 1]$$

On the other hand, you can easily verify that the set

$$D' = \left\{ \frac{a}{1} : a \in D \right\}$$

is a subring of Q isomorphic to D . In practice, we will identify D with D' .

THEOREM 4.40. Embedding Theorem.

*Let D be an integral domain, then there exists a field Q whose elements are fractions or quotients $\frac{a}{b}$, where a and $b \neq 0$ are in D . By identifying $a = \frac{a}{1}$ for all $a \in D$, we can regard D as a subring of Q . In such a case, every $a \neq 0$ in D has an inverse in Q , and each quotient in Q has the form $\frac{a}{b} = ab^{-1}$, where a and $b \neq 0$ are in D . The field Q is called the **field of fractions** of the integral domain D .*

PROOF. It remains to prove the last sentence. We have that

$$\frac{a}{b} = \frac{a}{1} \cdot \frac{1}{b} = \frac{a}{1} \left(\frac{b}{1} \right)^{-1} = ab^{-1},$$

after identifying $\frac{a}{1} = a$ and $\frac{b}{1} = b$. $\square$

4.8 Quaternions

We close the present chapter by introducing a new subset of “numbers”: the quaternions.

The set of complex numbers can be identified with points in the plane. Inspired by this fact, the Irish mathematician William Rowan Hamilton looked for the analogue algebraic structure to describe three-dimensional space. After a frustrating 10-year search, he finally realized that the desired algebraic structure must be four-dimensional and noncommutative. He called these new “numbers” **quaternions**.

Recall that complex numbers have the form $a + bi$ where a and b are real numbers and $i^2 = -1$. By analogy, the set $\mathbb{H}$ of quaternions is defined by

$$\mathbb{H} = \{a + bi + cj + dk \mid a, b, c, d \in \mathbb{R}\}.$$

Here, as for complex numbers we require that

$$a + bi + cj + dk = a' + b'i + c'j + d'k \Leftrightarrow a = a', b = b', c = c', d = d'.$$

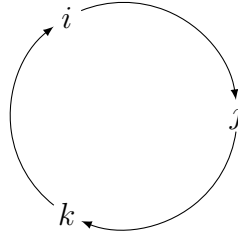
We also have that any real number commutes with each of i , j and k . With this, the multiplication in $\mathbb{H}$ is determined by the distributive laws once the products, i^2 , j^2 , ij , $\dots$ are specified. These in turn follow from the equations

$$i^2 = j^2 = k^2 = ijk = -1,$$

which yield the following formulas:

$$\begin{aligned} ij &= k = -ji \\ jk &= i = -kj \\ ki &= j = -ik \end{aligned}$$

These formulas are easy to remember from the diagram:



The product of any two of i , j and k taken clockwise around the circle is the next one; the product counterclockwise is the negative of the next one.

The ring of complex numbers $\mathbb{C}$ is a subring of $\mathbb{H}$ by identifying

$$a + bi = a + bi + 0j + 0k.$$

Let us see an example of how products in $\mathbb{H}$ are computed.

Example 4.41.

$$(3 - 4j)(2i + k) = 6i + 3k - 8ji - 4jk = 6i + 3k + 8k - 4i = 2i + 11k$$

If $z = a + bi$ is a complex number, we have that $\bar{z} = a - bi$ and $|z|^2 = a^2 + b^2$. The analogy between $\mathbb{C}$ and $\mathbb{H}$ leads to a natural extension of these important notions to $\mathbb{H}$. Given a quaternion $q = a + bi + cj + dk$, we define the **conjugate** q^* and the **norm** $N(q)$ as follows:

$$q^* = a - bi - cj - dk, \quad N(q) = a^2 + b^2 + c^2 + d^2.$$

It is easy to see that

$$qq^* = N(q) = q^*q,$$

for every quaternion q .

THEOREM 4.42. *The ring $\mathbb{H}$ is a noncommutative division ring. Moreover, if $q \neq 0$ in $\mathbb{H}$, then $q^{-1} = \frac{1}{N(q)}q^*$.*

PROOF. It is clear that $\mathbb{H}$ is noncommutative since, for example, $ij \neq ji$. If $q = a + bi + cj + dk \neq 0$ in $\mathbb{H}$, then one of a , b , c or d is nonzero, so $N(q) = a^2 + b^2 + c^2 + d^2$ is a nonzero real number. From the equations $qq^* = N(q) = q^*q$ we have that $q^{-1} = \frac{1}{N(q)}q^*$, finishing the proof. $\square$



Hope you've enjoyed the course!